



What is Statistics?

Statistics is a branch of mathematics that involves collecting, analysing, interpreting, and presenting data.

It provides tools and methods to understand and make sense of large amounts of data and to draw conclusions and make decisions based on the data.

In practice, statistics is used in a wide range of fields, such as business, economics, social sciences, medicine, and engineering. It is used to conduct research studies, analyse market trends, evaluate the effectiveness of treatments and interventions, and make forecasts and predictions.

Population Vs Sample

A **population** refers to the entire group of individuals, objects, or data points that a researcher is interested in studying. It includes all possible observations that could be made. For example, if you want to study the average height of all adults in a country, the population would be every single adult in that country.

A **sample** is a subset of the population. Because it is often impractical or impossible to study an entire population, researchers select a smaller, representative group (a sample) from which they collect data. The goal is that the findings from the sample can be generalized back to the larger population. For example, instead of measuring every adult's height, you might measure the height of a few thousand adults selected from different regions, age groups, etc., to form a sample.

Descriptive vs. Inferential Statistics

Feature	Descriptive Statistics	Inferential Statistics
Purpose	Summarize and describe characteristics of a dataset.	Make inferences and predictions about a population.
Scope	Focuses on known data (sample or population).	Extends findings from a sample to a larger population.
Output	Measures like mean, median, mode, standard deviation, frequencies, distributions.	Probability, hypothesis testing, confidence intervals, regression.
Questions	What <i>is</i> the data like?	What <i>can we conclude</i> about the population?
Example	The average height of students in a <i>classroom</i> .	Estimating the average height of <i>all adults in a country</i> based

Feature	Descriptive Statistics	Inferential Statistics
		on a sample.
Generalization	No generalization beyond the observed data.	Generalizes from sample to population.

Measures of Central Tendency

Measures of central tendency describe the central or typical value in a dataset. They provide a single value that represents the "center" of the distribution, helping us understand where most of the data points cluster.

Mean

The **mean** (or arithmetic mean/average) is the sum of all values divided by the number of values. It is the most commonly used measure of central tendency.

- **Formula (Population):** $\mu = \frac{\sum x_i}{N}$
- **Formula (Sample):** $\bar{x} = \frac{\sum x_i}{n}$
- **Use Case:** Best for symmetric distributions without outliers. Used extensively in statistical analysis.
- **Limitations:** Highly sensitive to outliers and extreme values. Can be misleading for skewed distributions.

Example: Calculating Mean

Dataset: Test scores of 10 students: 45, 52, 58, 63, 65, 70, 72, 78, 85, 90

Step 1: Sum all values $\sum x_i = 45 + 52 + 58 + 63 + 65 + 70 + 72 + 78 + 85 + 90 = 678$

Step 2: Count the number of values $n = 10$

Step 3: Divide sum by count $\bar{x} = \frac{678}{10} = 67.8$

Mean = 67.8

Interpretation: The average test score is 67.8 points.

Median

The **median** is the middle value when data is arranged in order. It divides the dataset into two equal halves, with 50% of values below and 50% above.

- **Formula:**
 - If n is odd: Median = value at position $\frac{n+1}{2}$
 - If n is even: Median = average of values at positions $\frac{n}{2}$ and $\frac{n}{2} + 1$
- **Use Case:** Best for skewed distributions or when outliers are present. Used for income data, housing prices, etc.
- **Advantages:** Not affected by outliers or extreme values. More representative for skewed data.

Example: Calculating Median (Even Number of Values)

Dataset: Test scores of 10 students: 45, 52, 58, 63, 65, 70, 72, 78, 85, 90

Step 1: Arrange data in ascending order (already arranged) 45, 52, 58, 63, 65, 70, 72, 78, 85, 90

Step 2: Count the number of values $n = 10$ (even)

Step 3: Find the two middle values

- Position $\frac{n}{2} = \frac{10}{2} = 5 \rightarrow$ Value = 65
- Position $\frac{n}{2} + 1 = 5 + 1 = 6 \rightarrow$ Value = 70

Step 4: Calculate average of the two middle values $\text{Median} = \frac{65 + 70}{2} = \frac{135}{2} = 67.5$

Median = 67.5

Interpretation: Half of the students scored below 67.5, and half scored above 67.5.

Example: Calculating Median (Odd Number of Values)

Dataset: Test scores of 9 students: 45, 52, 58, 63, 65, 72, 78, 85, 90

Step 1: Arrange data in ascending order (already arranged) 45, 52, 58, 63, 65, 72, 78, 85, 90

Step 2: Count the number of values $n = 9$ (odd)

Step 3: Find the middle position $\text{Position} = \frac{n+1}{2} = \frac{9+1}{2} = 5$

Step 4: The value at position 5 is the median $\text{Median} = 65$

Median = 65

Mode

The **mode** is the value that appears most frequently in a dataset. A dataset can have one mode (unimodal), two modes (bimodal), multiple modes (multimodal), or no mode at all.

- **Formula:** No mathematical formula; simply the most frequent value(s)
- **Use Case:** Best for categorical data. Useful for identifying the most common value in any type of data.
- **Characteristics:**
 - Can be used with nominal data (e.g., colors, categories)
 - Can have multiple modes or no mode
 - Not affected by outliers

Example: Calculating Mode

Dataset 1: Test scores: 45, 52, 58, 63, 65, 70, 70, 72, 78, 85, 90

Step 1: Count the frequency of each value

Value	Frequency
45	1
52	1
58	1
63	1
65	1
70	2
72	1
78	1
85	1

Value	Frequency
90	1

Step 2: Identify the value(s) with the highest frequency $\text{Mode} = 70$ (appears 2 times)

Mode = 70

Example: Bimodal Distribution

Dataset 2: Test scores: 45, 52, 58, 58, 63, 65, 70, 72, 72, 78, 85

Frequency count: 58 appears 2 times, 72 appears 2 times

Modes = 58 and 72 (bimodal distribution)

Example: No Mode

Dataset 3: Test scores: 45, 52, 58, 63, 65, 70, 72, 78, 85, 90

All values appear exactly once

Mode = None (no mode exists)

Summary of Results for Our Example

Dataset: 45, 52, 58, 63, 65, 70, 72, 78, 85, 90

Measure	Value	Interpretation
Mean	67.8	Average score
Median	67.5	Middle score
Mode	None	No repeating value

Effect of Outliers

Original Dataset: 45, 52, 58, 63, 65, 70, 72, 78, 85, 90

Modified Dataset (with outlier): 45, 52, 58, 63, 65, 70, 72, 78, 85, **200**

Measure	Original	With Outlier	Change
Mean	67.8	78.8	+11.0 (16.2% increase)
Median	67.5	67.5	0 (no change)
Mode	None	None	no change

Key Insight: The median is **robust to outliers**, while the mean is **highly sensitive**.

Relationship Between Mean, Median, and Mode

The relationship between these measures reveals the shape of the distribution:

Symmetric Distribution

- **Mean = Median = Mode**
- Example: Normal distribution (bell curve)

Right-Skewed Distribution (Positively Skewed)

- **Mode < Median < Mean**
- The mean is pulled toward the right tail
- Example: Income distribution, housing prices

Left-Skewed Distribution (Negatively Skewed)

- **Mean < Median < Mode**
- The mean is pulled toward the left tail
- Example: Age at retirement, exam scores when most students do well

Comparison of Measures of Central Tendency

Measure	Formula	Best For	Advantages	Disadvantages
Mean	$\frac{\sum x_i}{n}$	Symmetric distributions without outliers	Uses all data; mathematically tractable; widely used	Sensitive to outliers; can be misleading for skewed data
Median	Middle value when ordered	Skewed distributions; data with outliers	Robust to outliers; represents true center for skewed data	Ignores magnitude of values; less mathematically tractable

Measure	Formula	Best For	Advantages	Disadvantages
Mode	Most frequent value	Categorical data; finding most common value	Can be used with any data type; easy to understand	May not exist or may not be unique; doesn't use all data

Decision Guide: Which Measure to Use?

Use the **MEAN** when:

- Data is symmetric and normally distributed
- No significant outliers are present
- You need to use the data in further calculations
- Working with interval or ratio data

Use the **MEDIAN** when:

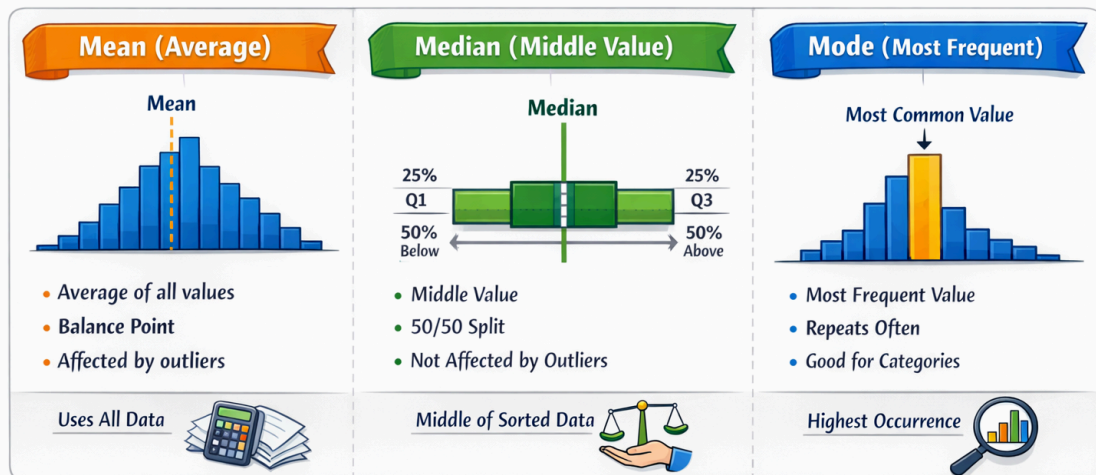
- Data is skewed
- Outliers are present
- You want the "typical" middle value
- Working with ordinal, interval, or ratio data
- Reporting income, housing prices, or similar data

Use the **MODE** when:

- Working with categorical/nominal data
- You want to know the most common value
- Data is discrete with repeating values
- Describing the most popular choice

Key Insights:

- **Mean** is the most commonly used measure but is sensitive to extreme values.
- **Median** provides a better measure of central tendency when dealing with skewed distributions or outliers.
- **Mode** is useful for categorical data and identifying the most frequent value, but may not always exist or be unique.
- When choosing a measure, consider whether your data contains outliers and the shape of your distribution.



Mean • Median • Mode in Data Analysis

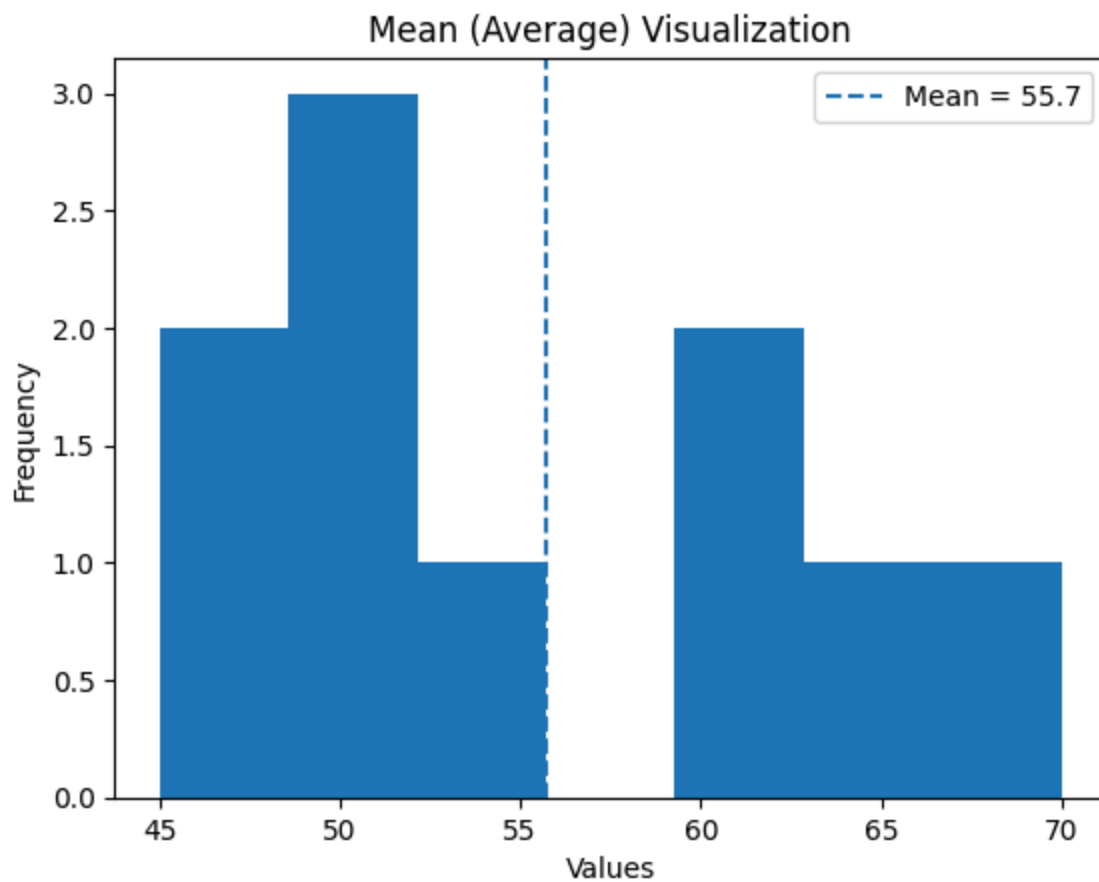
Measures of Central Tendency

```
In [ ]: import numpy as np
import matplotlib.pyplot as plt

data = np.array([45, 48, 50, 50, 52, 55, 60, 62, 65, 70])
mean = np.mean(data)

plt.hist(data, bins=7)
plt.axvline(mean, linestyle='--', label=f"Mean = {mean}")

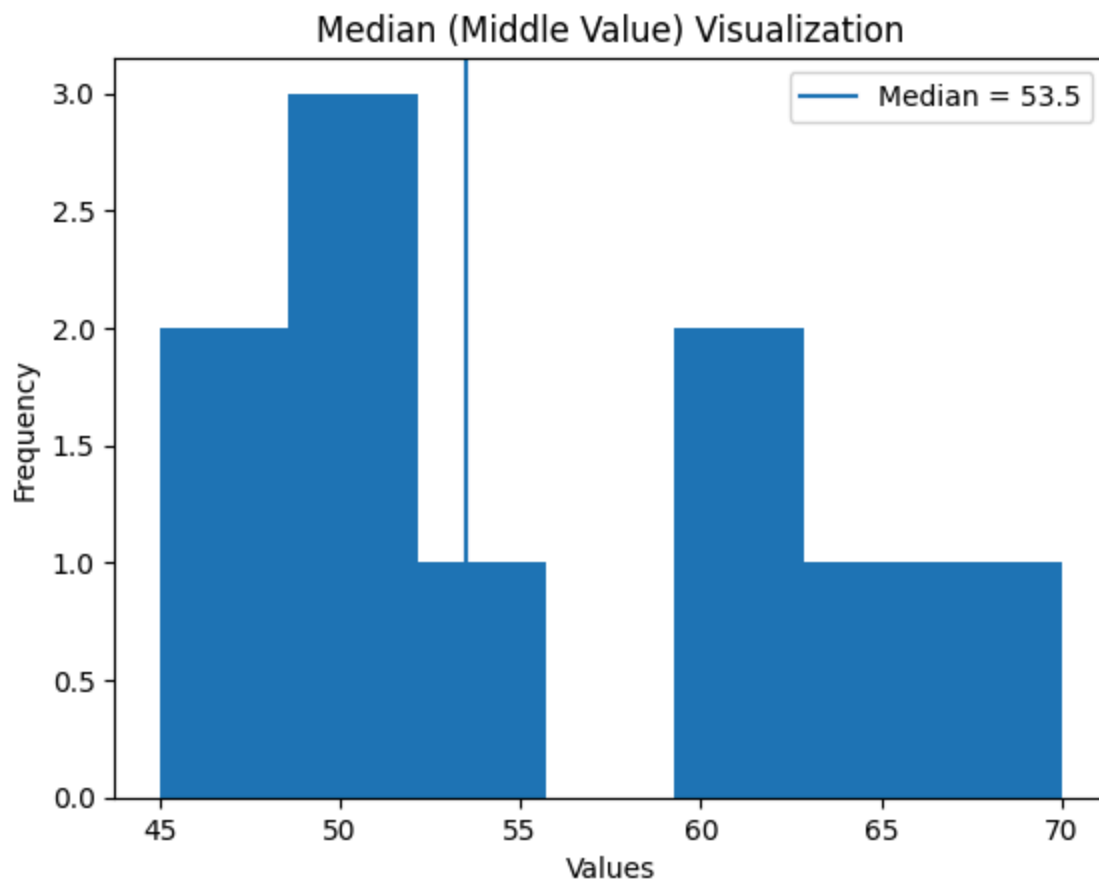
plt.title("Mean (Average) Visualization")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```

```
In [ ]: median = np.median(data)

plt.hist(data, bins=7)
plt.axvline(median, linestyle='-', label=f"Median = {median}")

plt.title("Median (Middle Value) Visualization")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```

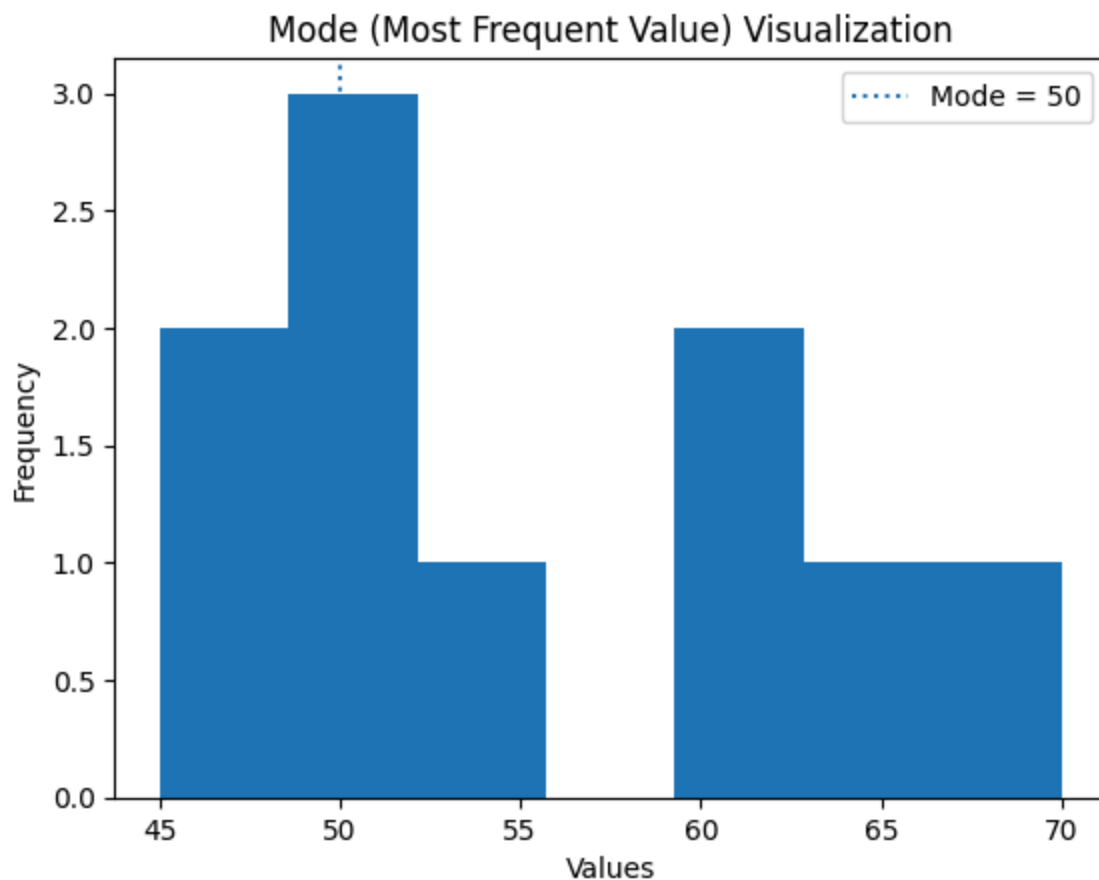


```
In [ ]: from scipy import stats

mode = stats.mode(data, keepdims=True)[0][0]

plt.hist(data, bins=7)
plt.axvline(mode, linestyle=':', label=f"Mode = {mode}")

plt.title("Mode (Most Frequent Value) Visualization")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```



```
In [ ]: import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# Sample data
data = np.array([45, 48, 50, 50, 52, 55, 60, 62, 65, 70])

mean = np.mean(data)
median = np.median(data)
mode = stats.mode(data, keepdims=True)[0][0]

# -----
# 1. Histogram (Mean, Median, Mode)
# -----
plt.hist(data, bins=7, alpha=0.7)
plt.axvline(mean, linestyle='--', label=f"Mean = {mean}")
plt.axvline(median, linestyle='-', label=f"Median = {median}")
plt.axvline(mode, linestyle=':', label=f"Mode = {mode}")

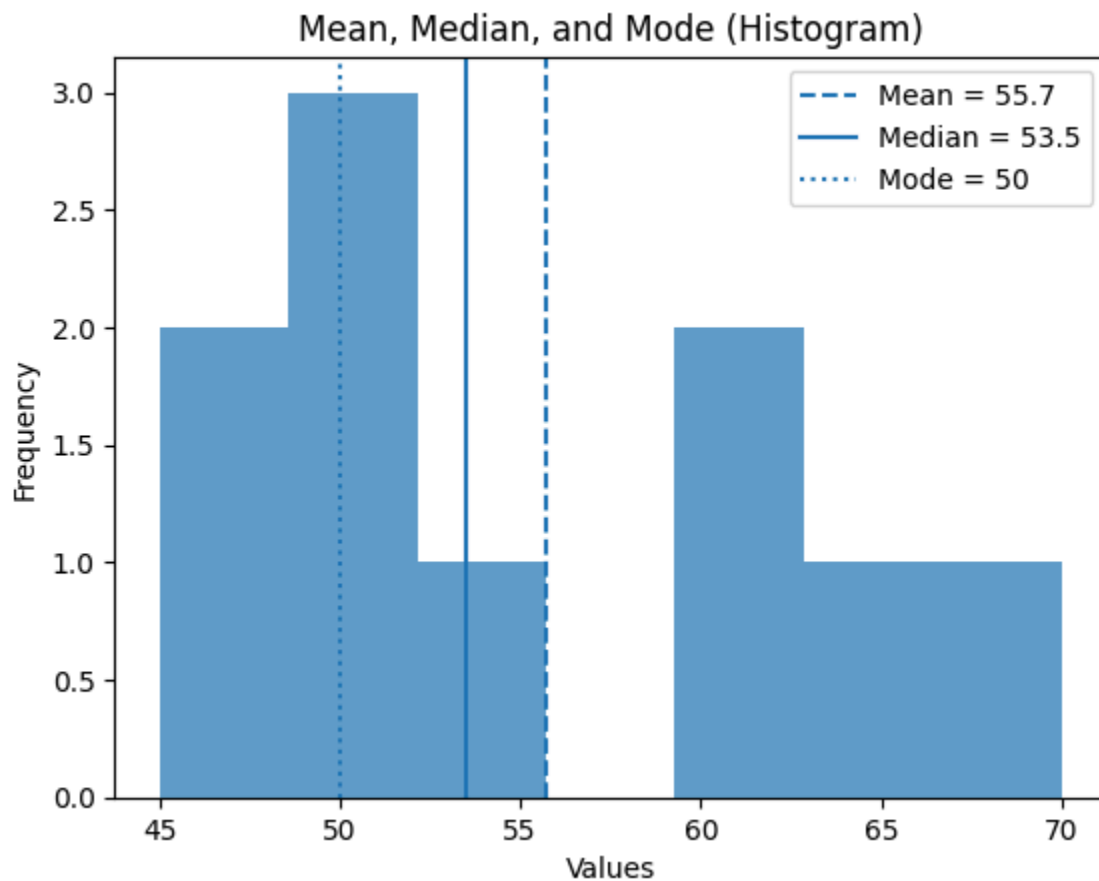
plt.title("Mean, Median, and Mode (Histogram)")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.legend()
plt.show()

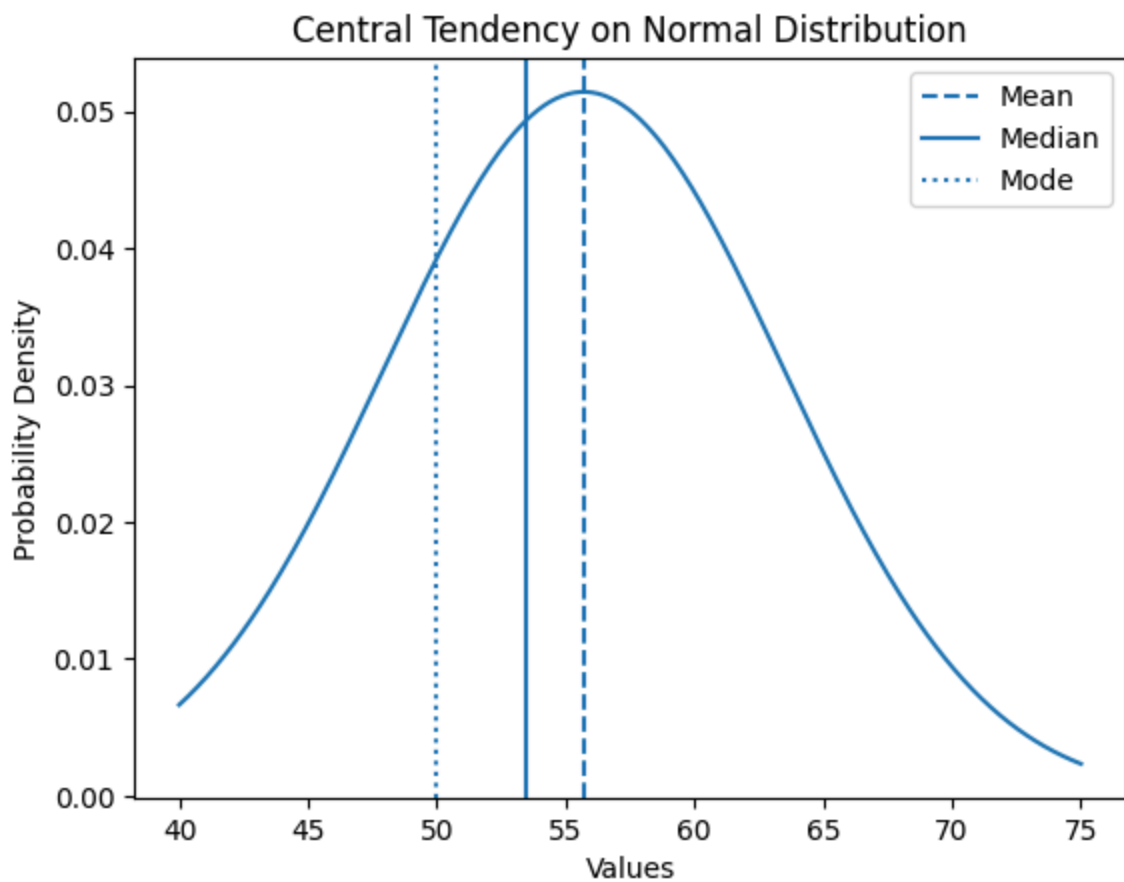
# -----
```

```
# 3. Normal Distribution with Mean, Median, Mode
# -----
x = np.linspace(40, 75, 300)
y = stats.norm.pdf(x, mean, np.std(data))

plt.plot(x, y)
plt.axvline(mean, linestyle='--', label="Mean")
plt.axvline(median, linestyle='-', label="Median")
plt.axvline(mode, linestyle=':', label="Mode")

plt.title("Central Tendency on Normal Distribution")
plt.xlabel("Values")
plt.ylabel("Probability Density")
plt.legend()
plt.show()
```





```
In [ ]: #Real World Data
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [ ]: df1=pd.read_csv('/content/insurance.csv')
```

```
In [ ]: df1.head()
```

```
Out[ ]:
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
In [ ]: df1["age"]
```

Out[]: **age**

0	19
1	18
2	28
3	33
4	32
...	...
1333	50
1334	18
1335	18
1336	21
1337	61

1338 rows × 1 columns

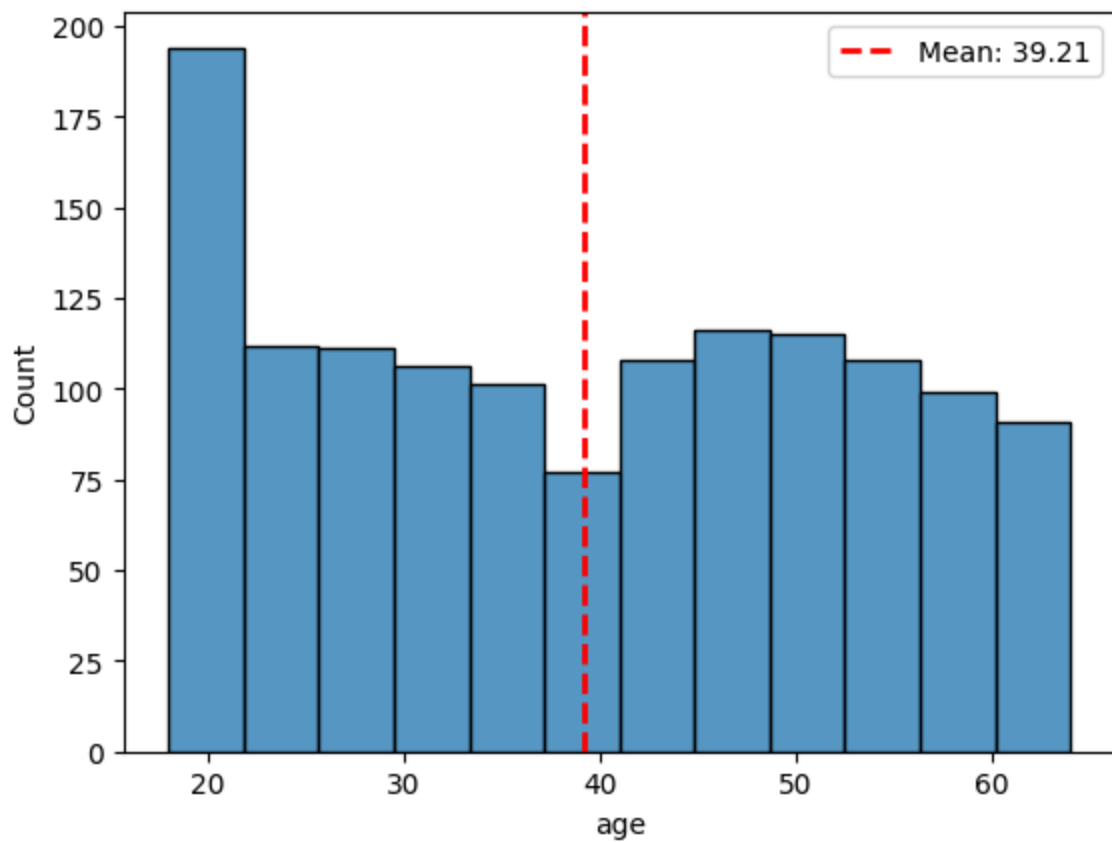
dtype: int64

```
In [ ]: #Measure of Central Tendency (Median, Mean, Mode)
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Calculate mean
mean = np.mean(df1['age'])

# Create histogram
sns.histplot(x="age", data=df1)

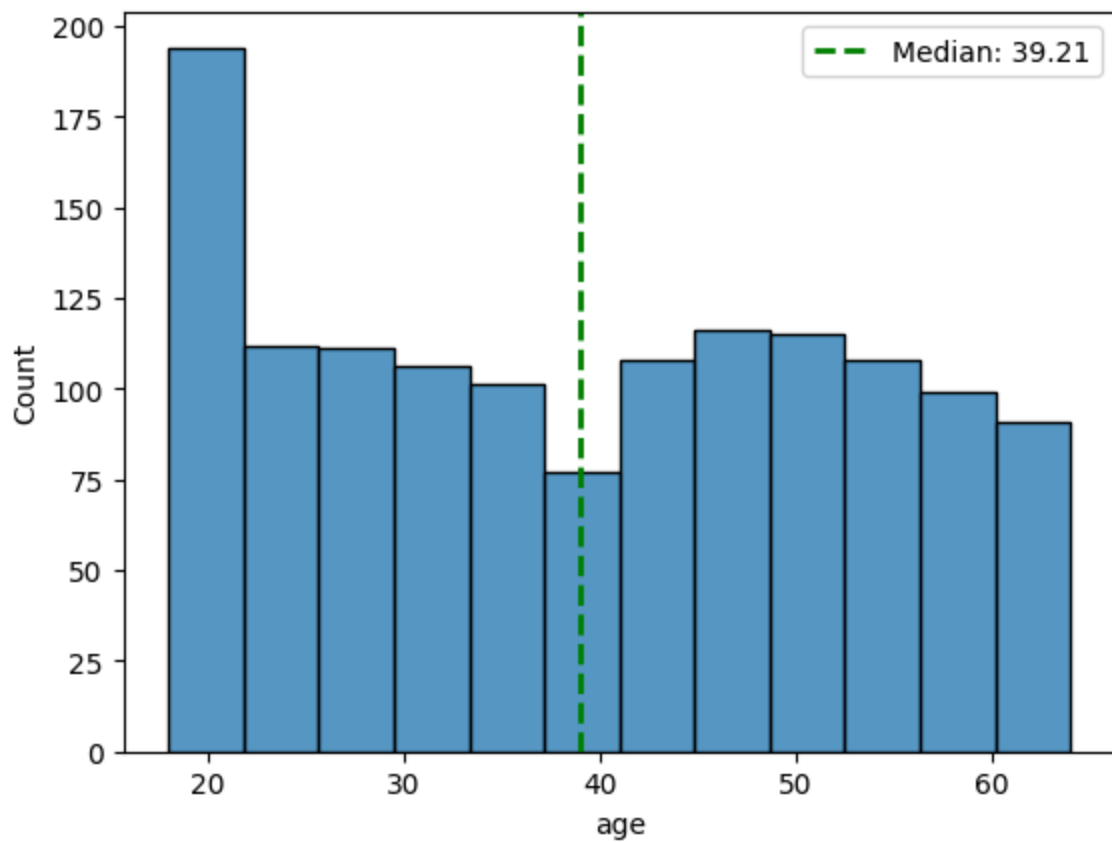
# Add vertical line at mean
plt.axvline(x=mean, color='red', linestyle='--', linewidth=2, label=f'Mean: {mean}')
plt.legend()
plt.show()
```



```
In [ ]: # Calculate median
median = np.median(df1['age'])

# Create histogram
sns.histplot(x="age", data=df1)

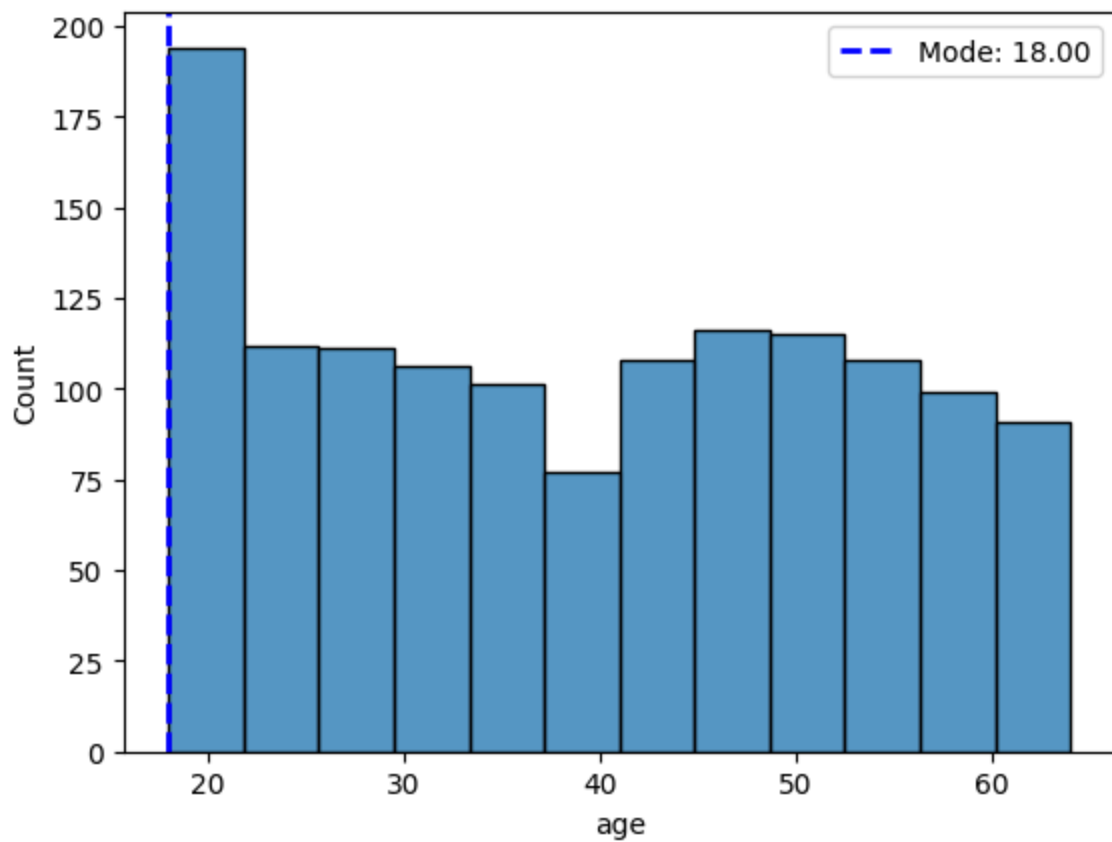
# Add vertical line at median
plt.axvline(x=median, color='Green', linestyle='--', linewidth=2, label=f'Median: {median}')
plt.legend()
plt.show()
```



```
In [ ]: # Calculate mode
mode = df1['age'].mode()[0]

# Create histogram
sns.histplot(x="age", data=df1)

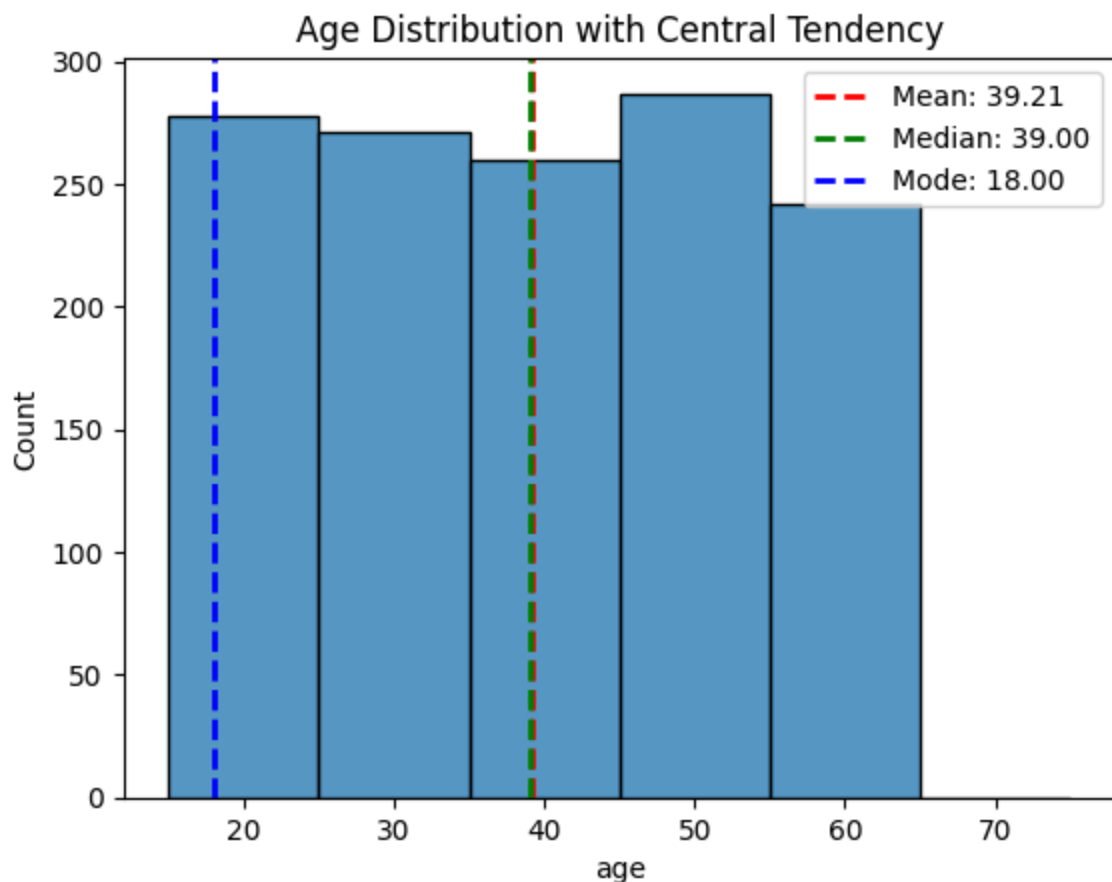
# Add vertical line at mode
plt.axvline(x=mode, color='Blue', linestyle='--', linewidth=2, label=f'Mode: {mode}')
plt.legend()
plt.show()
```

```
In [ ]: # Create histogram together mean median and mode
sns.histplot(x="age", data=df1, bins=[15, 25, 35, 45, 55, 65, 75])

# Add vertical lines at mean, median and mode
plt.axvline(x=mean, color='red', linestyle='--', linewidth=2, label=f'Mean: {mean}')
plt.axvline(x=median, color='green', linestyle='--', linewidth=2, label=f'Median: {median}')
plt.axvline(x=mode, color='blue', linestyle='--', linewidth=2, label=f'Mode: {mode}')

plt.legend()
plt.title('Age Distribution with Central Tendency')
plt.show()
```



Measures of Variability

Measures of variability (or dispersion) describe how spread out the data points are in a dataset. They provide information about the width of the distribution, in contrast to measures of central tendency which describe the central value.

Range

The **range** is the simplest measure of variability. It is the difference between the highest and lowest values in a dataset.

- **Formula:** $\text{Range} = \text{Maximum Value} - \text{Minimum Value}$
- **Use Case:** Quick and easy to calculate, but highly sensitive to outliers.
- **Limitations:** Only considers the two extreme values, ignoring the distribution of data in between.

Example: Calculating Range

Dataset: Test scores of 10 students: 45, 52, 58, 63, 65, 70, 72, 78, 85, 90

Step 1: Identify the maximum value = 90

Step 2: Identify the minimum value = 45

Step 3: Calculate Range = 90 - 45 = **45**

Variance

The **variance** measures the average squared deviation of each data point from the mean. It gives more weight to outliers because it squares the differences.

- **Formula (Population):** $\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$
- **Formula (Sample):** $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$
- **Use Case:** Provides a good understanding of data spread. Used in many statistical tests.
- **Limitations:** Units are squared, making it less intuitive to interpret than standard deviation.

Example: Calculating Sample Variance

Dataset: Test scores of 10 students: 45, 52, 58, 63, 65, 70, 72, 78, 85, 90

Step 1: Calculate the mean $\bar{x} = \frac{45 + 52 + 58 + 63 + 65 + 70 + 72 + 78 + 85 + 90}{10} = \frac{678}{10} = 67.8$

Step 2: Calculate deviations from mean and square them

Value (\$x_i\$)	Deviation (\$x_i - \bar{x}\$)	Squared Deviation \$(x_i - \bar{x})^2\$
45	45 - 67.8 = -22.8	519.84
52	52 - 67.8 = -15.8	249.64
58	58 - 67.8 = -9.8	96.04
63	63 - 67.8 = -4.8	23.04
65	65 - 67.8 = -2.8	7.84
70	70 - 67.8 = 2.2	4.84
72	72 - 67.8 = 4.2	17.64
78	78 - 67.8 = 10.2	104.04
85	85 - 67.8 = 17.2	295.84
90	90 - 67.8 = 22.2	492.84

Step 3: Sum the squared deviations $\sum (x_i - \bar{x})^2 = 1811.60$

Step 4: Divide by (n-1) for sample variance $s^2 = \frac{1811.60}{10-1} = \frac{1811.60}{9} = 201.29$

Sample Variance = 201.29

Standard Deviation

The **standard deviation** is the square root of the variance. It is the most commonly used measure of variability because it is expressed in the same units as the data, making it easier to interpret.

- **Formula (Population):** $\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$
- **Formula (Sample):** $s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$
- **Use Case:** Widely used to describe the typical distance of data points from the mean. Useful for comparing variability between different datasets.
- **Sensitivity:** Sensitive to outliers, but less so than the range.

Example: Calculating Sample Standard Deviation

Dataset: Test scores of 10 students: 45, 52, 58, 63, 65, 70, 72, 78, 85, 90

Step 1: We already calculated the sample variance = 201.29

Step 2: Take the square root of variance $s = \sqrt{201.29} = 14.19$

Sample Standard Deviation = 14.19

Interpretation: On average, test scores deviate from the mean (67.8) by about 14.19 points.

Mean Absolute Deviation (MAD)

The **Mean Absolute Deviation (MAD)** measures the average absolute deviation of each data point from the mean. Unlike variance and standard deviation, it does not square the deviations, making it less sensitive to outliers.

- **Formula (Population):** $MAD = \frac{\sum |x_i - \mu|}{N}$
- **Formula (Sample):** $MAD = \frac{\sum |x_i - \bar{x}|}{n}$

- **Use Case:** Useful when you want a measure of spread that is robust to outliers and easy to interpret.
- **Limitations:** Less commonly used in advanced statistical procedures compared to standard deviation. Not as mathematically tractable for theoretical work.

Example: Calculating MAD

Dataset: Test scores of 10 students: 45, 52, 58, 63, 65, 70, 72, 78, 85, 90

Step 1: Mean = 67.8 (already calculated)

Step 2: Calculate absolute deviations from mean

Value (\$x_i\$)	Deviation (\$x_i - \bar{x}\$)	Absolute Deviation \$ x_i - \bar{x} \$	
45	-22.8	22.8	
52	-15.8	15.8	
58	-9.8	9.8	
63	-4.8	4.8	
65	-2.8	2.8	
70	2.2	2.2	
72	4.2	4.2	
78	10.2	10.2	
85	17.2	17.2	
90	22.2	22.2	

Step 3: Sum the absolute deviations $\sum |x_i - \bar{x}| = 112.0$

Step 4: Divide by n $MAD = \frac{112.0}{10} = 11.2$

MAD = 11.2

Interpretation: On average, test scores deviate from the mean by 11.2 points (without squaring).

Summary of Results for Our Example

Dataset: 45, 52, 58, 63, 65, 70, 72, 78, 85, 90

Measure	Value	Units
Mean	67.8	points
Range	45	points
Variance	201.29	points ²
Standard Deviation	14.19	points
MAD	11.2	points

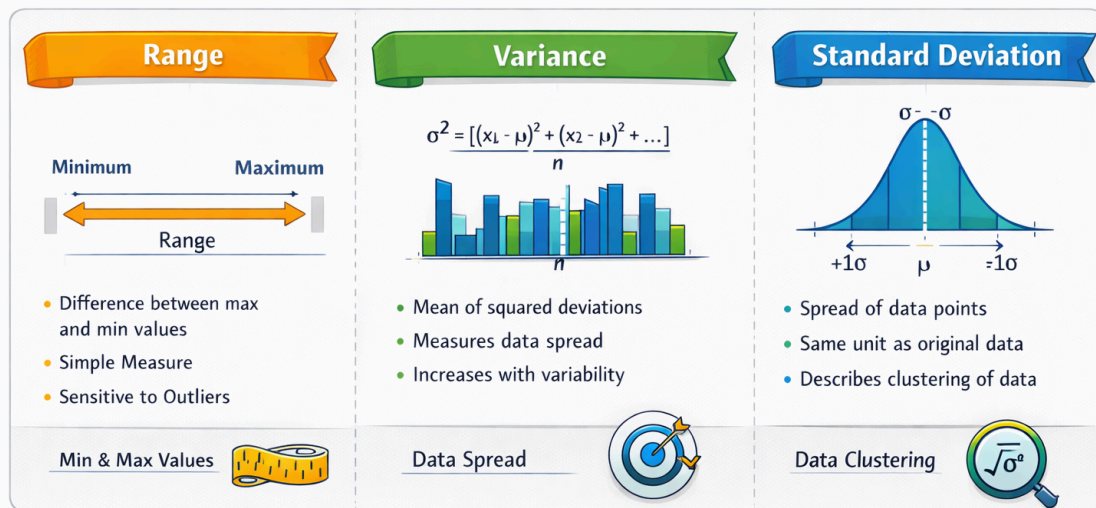
Notice: MAD (11.2) is smaller than Standard Deviation (14.19) because squaring deviations in SD gives more weight to larger deviations.

Comparison of Measures of Variability

Measure Formula Units Sensitivity to Outliers Advantages Disadvantages
----- ----- ----- ----- ----- -----
Same as data Very High Simple and quick to calculate Only uses two values; ignores data distribution Range Max - Min
Squared units High Foundation for many statistical tests; uses all data points Difficult to interpret due to squared units Variance $\frac{\sum (x_i - \bar{x})^2}{n-1}$
$\sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$ Same as data High Widely used; same units as data; easier to interpret than variance Sensitive to outliers Standard Deviation
$\frac{\sum x_i - \bar{x} }{n}$ Same as data Moderate Robust to outliers; easy to interpret; uses all data points Less common in statistical tests; less mathematically convenient MAD

Key Insights:

- **Range** is the quickest to calculate but provides the least information about the distribution.
- **Variance and Standard Deviation** are most commonly used in inferential statistics and provide comprehensive information about spread, but are sensitive to extreme values.
- **MAD** offers a good middle ground when dealing with datasets that contain outliers, providing a more robust measure of typical deviation.
- When choosing a measure, consider whether your data contains outliers and what you need the measure for (descriptive purposes vs. inferential statistics).



Range • Variance • Standard Deviation in Data Analysis

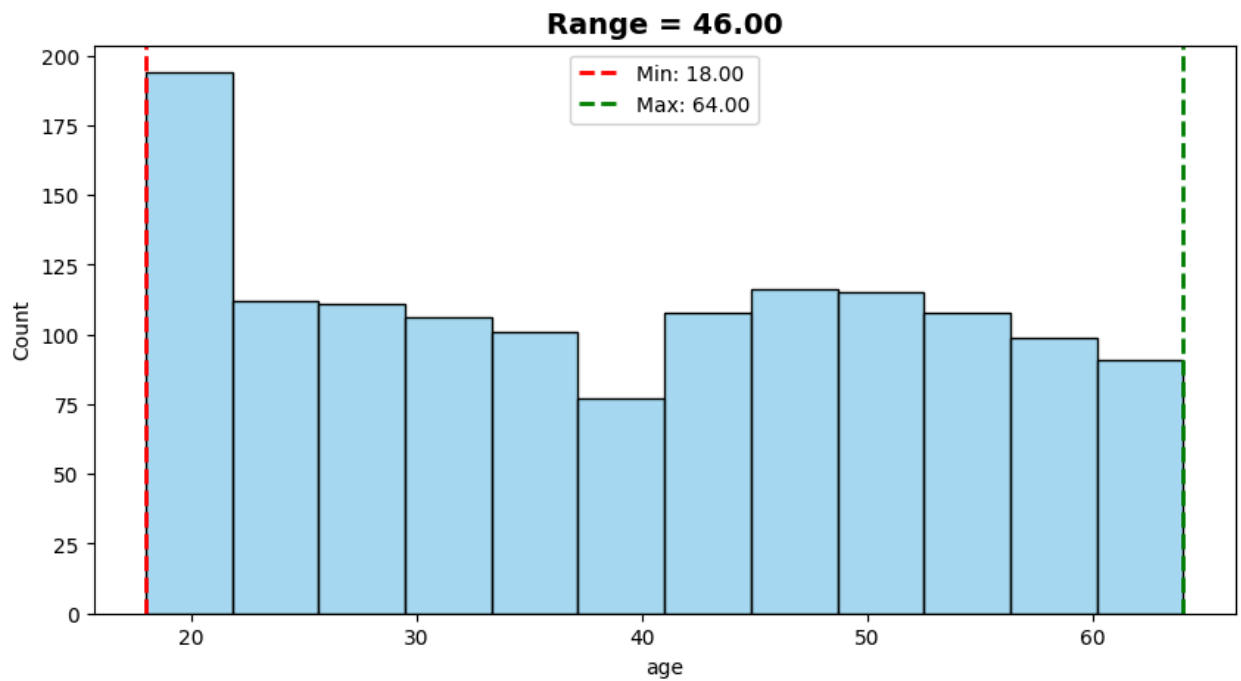
— Measures of Variability —

In []: *#Measures of Variability*

```
#1.Range
maxr=df1['age'].max()
minr=df1['age'].min()
range=maxr-minr
print("Range:",range)
```

Range: 46

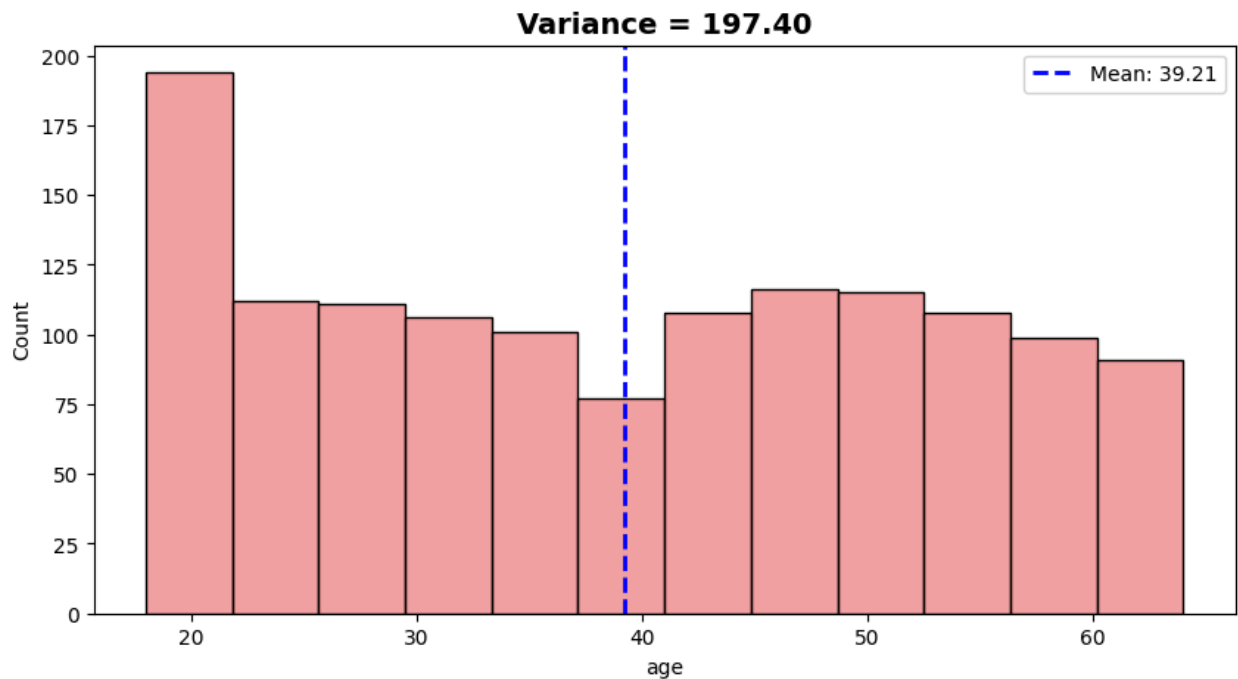
```
# 1. Range Visualization
plt.figure(figsize=(10, 5))
sns.histplot(x="age", data=df1, color='skyblue')
plt.axvline(x=df1['age'].min(), color='red', linestyle='--', linewidth=2, label='Min')
plt.axvline(x=df1['age'].max(), color='green', linestyle='--', linewidth=2, label='Max')
plt.title(f'Range = {range:.2f}', fontweight='bold', fontsize=14)
plt.legend()
plt.show()
```



```
In [ ]: #2. Varienve
var=df1['age'].var()
print("Variance:",var)
```

Variance: 197.40138665754424

```
In [ ]: # 2. Variance Visualization
plt.figure(figsize=(10, 5))
sns.histplot(x="age", data=df1, color='lightcoral')
plt.axvline(x=mean, color='blue', linestyle='--', linewidth=2, label=f'Mean: {
plt.title(f'Variance = {var:.2f}', fontweight='bold', fontsize=14)
plt.legend()
plt.show()
```

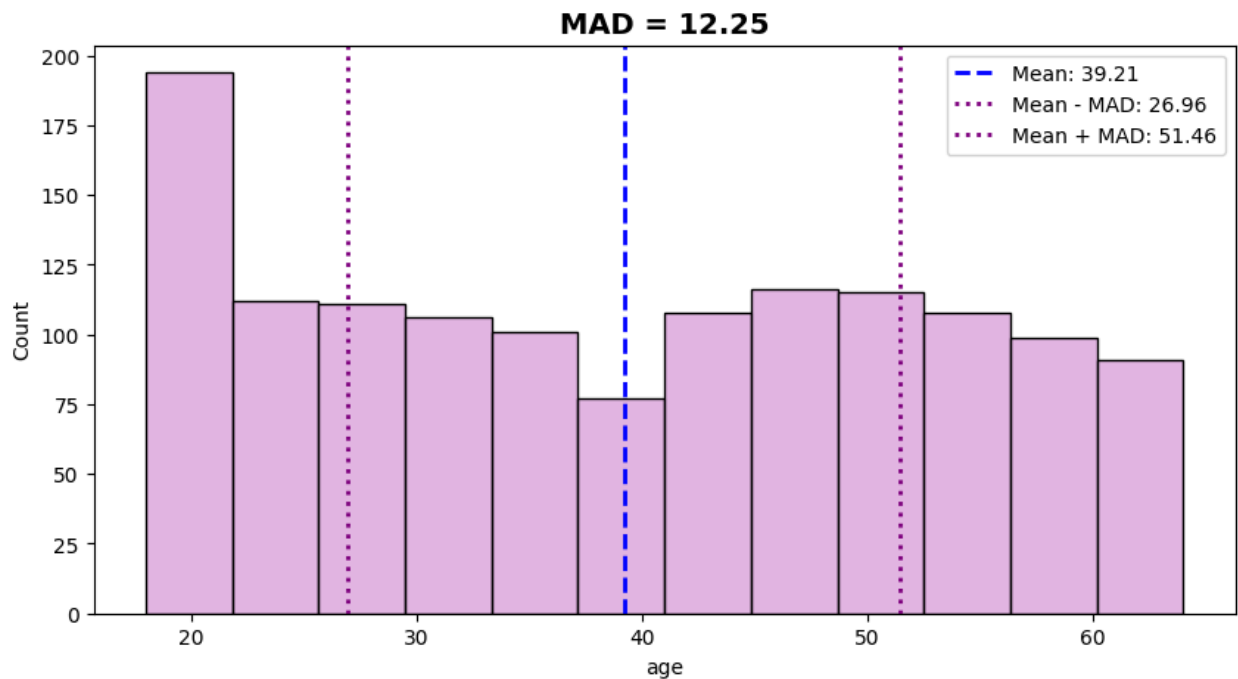



```
In [ ]: #3. Mean Absolute Deviation
# Method 1: Using your approach (correct)
mad = np.mean(np.abs(df1['age'] - df1['age'].mean()))

# Method 2: Step by step
mean_age = df1['age'].mean()
deviations = df1['age'] - mean_age
absolute_deviations = np.abs(deviations)
mad = np.mean(absolute_deviations)
print("Mean Absolute Deviation:", round(mad, 2))
```

Mean Absolute Deviation: 12.25

```
In [ ]: # 4. MAD Visualization
plt.figure(figsize=(10, 5))
sns.histplot(x="age", data=df1, color='plum')
plt.axvline(x=mean_age, color='blue', linestyle='--', linewidth=2, label=f'Mean: {mean_age:.2f}')
plt.axvline(x=mean_age - mad, color='purple', linestyle=':', linewidth=2, label=f'MAD - {mad:.2f}')
plt.axvline(x=mean_age + mad, color='purple', linestyle=':', linewidth=2, label=f'MAD + {mad:.2f}')
plt.title(f'MAD = {mad:.2f}', fontweight='bold', fontsize=14)
plt.legend()
plt.show()
```



In []: **#4. Standard Deviation**

```
# Method 1: Using NumPy (sample standard deviation)
std = np.std(df1['age'])

# Method 2: Using pandas
std = df1['age'].std()

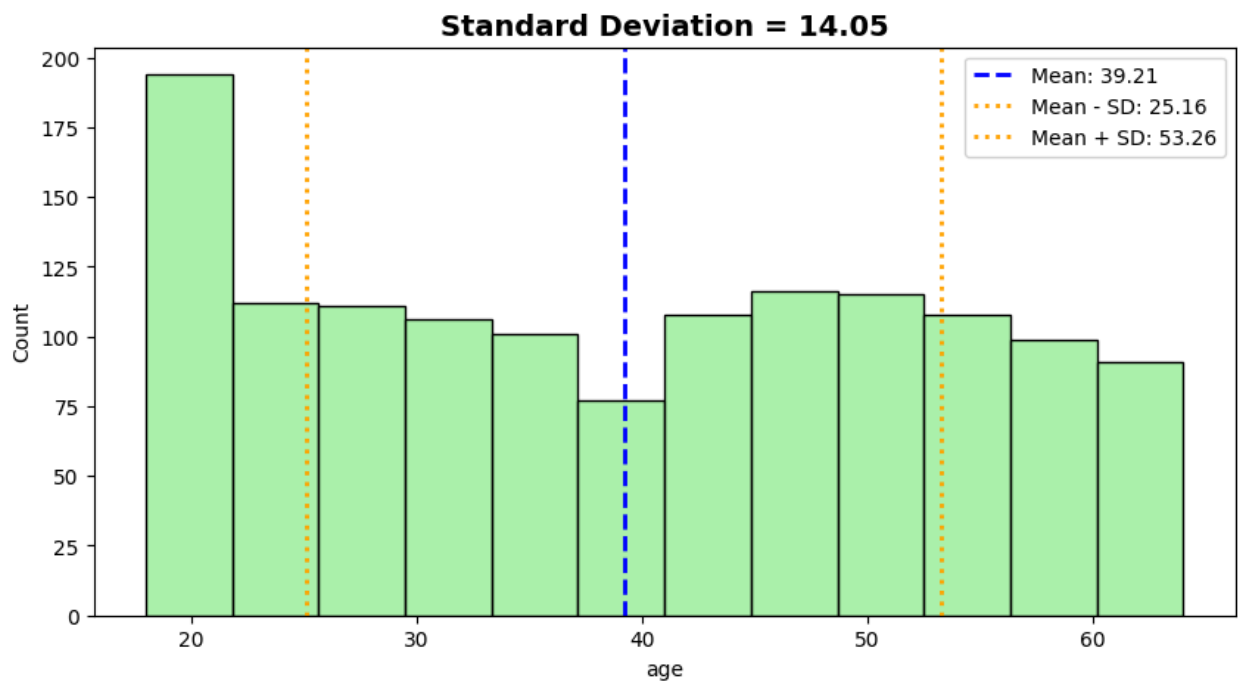
# Method 3: Step by step from variance
variance = df1['age'].var()
std = np.sqrt(variance)

print(f"Standard Deviation: {std:.2f}")
```

Standard Deviation: 14.05

In []: **# 3. Standard Deviation Visualization**

```
plt.figure(figsize=(10, 5))
sns.histplot(x="age", data=df1, color='lightgreen')
plt.axvline(x=mean_age, color='blue', linestyle='--', linewidth=2, label=f'Mean: {mean_age:.2f}')
plt.axvline(x=mean_age - std, color='orange', linestyle=':', linewidth=2, label=f'Mean - SD: {mean_age - std:.2f}')
plt.axvline(x=mean_age + std, color='orange', linestyle=':', linewidth=2, label=f'Mean + SD: {mean_age + std:.2f}')
plt.title(f'Standard Deviation = {std:.2f}', fontweight='bold', fontsize=14)
plt.legend()
plt.show()
```



Percentiles, Percentages, Outliers, and Box Plot

This document covers important statistical concepts for understanding data distribution, identifying unusual values, and visualizing data spread.

Percentiles

A **percentile** is a value below which a certain percentage of data falls. For example, the 75th percentile is the value below which 75% of the data lies.

- **Definition:** The Pth percentile is a value such that P% of the data is less than or equal to this value.
- **Common Percentiles:**
 - 25th percentile (Q1): First Quartile
 - 50th percentile (Q2): Median
 - 75th percentile (Q3): Third Quartile
- **Use Case:** Understanding relative standing in a dataset, comparing test scores, analyzing salary distributions.

Formula for Calculating Percentiles

Position Formula: $L = \frac{P}{100} \times (n + 1)$

Where:

- L = Position of the percentile
- P = Desired percentile (0 to 100)
- n = Number of data points

If L is a whole number: The percentile is the value at position L

If L is not a whole number: Interpolate between the two surrounding values

Example: Calculating Percentiles

Dataset: Test scores of 12 students (sorted): 45, 52, 58, 63, 65, 70, 72, 75, 78, 85, 90, 95

Find the 25th percentile (Q1), 50th percentile (Q2/Median), and 75th percentile (Q3)

Calculating 25th Percentile (Q1)

Step 1: Count the number of values $n = 12$

Step 2: Calculate the position $L = \frac{25}{100} \times (12 + 1) = 0.25 \times 13 = 3.25$

Step 3: Since $L = 3.25$ is not a whole number, interpolate between position 3 and position 4

- Value at position 3 = 58
- Value at position 4 = 63

Step 4: Interpolate $Q1 = 58 + 0.25 \times (63 - 58) = 58 + 0.25 \times 5 = 58 + 1.25 = 59.25$

25th Percentile (Q1) = 59.25

Interpretation: 25% of students scored below 59.25.

Calculating 50th Percentile (Q2/Median)

Step 1: Calculate the position $L = \frac{50}{100} \times (12 + 1) = 0.50 \times 13 = 6.5$

$$13 = 6.5$$

Step 2: Interpolate between position 6 and position 7

- Value at position 6 = 70
- Value at position 7 = 72

Step 3: Interpolate $Q2 = 70 + 0.5 \times (72 - 70) = 70 + 0.5 \times 2 = 70 + 1 = 71$

50th Percentile (Q2/Median) = 71

Interpretation: 50% of students scored below 71 (this is the median).

Calculating 75th Percentile (Q3)

Step 1: Calculate the position $L = \frac{75}{100} \times (12 + 1) = 0.75 \times 13 = 9.75$

Step 2: Interpolate between position 9 and position 10

- Value at position 9 = 78
- Value at position 10 = 85

Step 3: Interpolate $Q3 = 78 + 0.75 \times (85 - 78) = 78 + 0.75 \times 7 = 78 + 5.25 = 83.25$

75th Percentile (Q3) = 83.25

Interpretation: 75% of students scored below 83.25.

Summary of Percentiles

Percentile	Value	Interpretation
Q1 (25th)	59.25	25% scored below this
Q2 (50th)	71	50% scored below this (median)
Q3 (75th)	83.25	75% scored below this

Percentages

A **percentage** represents a part of a whole, expressed as a fraction of 100. Percentages are used to compare proportions and express relative quantities.

- **Formula:** $\text{Percentage} = \frac{\text{Part}}{\text{Whole}} \times 100\%$
- **Use Case:** Expressing proportions, calculating growth rates, analyzing compositions.

Example 1: Basic Percentage Calculation

Problem: A student answered 42 questions correctly out of 50 total questions. What percentage did they get correct?

Step 1: Identify the part and whole

- Part = 42 (correct answers)
- Whole = 50 (total questions)

Step 2: Apply the formula $\text{Percentage} = \frac{42}{50} \times 100\% = 0.84 \times 100\% = 84\%$

Answer: 84%

Interpretation: The student answered 84% of the questions correctly.

Example 2: Finding a Value from a Percentage

Problem: If 65% of students passed an exam and there are 200 students total, how many students passed?

Step 1: Identify the percentage and whole

- Percentage = 65%
- Whole = 200 students

Step 2: Convert percentage to decimal $65\% = \frac{65}{100} = 0.65$

Step 3: Calculate the part $\text{Part} = 0.65 \times 200 = 130$

Answer: 130 students passed

Example 3: Percentage Change

Problem: A product's price increased from \$80 to \$100. What is the percentage increase?

Formula for Percentage Change: $\text{Percentage Change} = \frac{\text{New Value} - \text{Old Value}}{\text{Old Value}} \times 100\%$

Step 1: Identify old and new values

- Old Value = \$80
- New Value = \$100

Step 2: Calculate the change $\text{Change} = 100 - 80 = 20$

Step 3: Calculate percentage change $\text{Percentage Change} = \frac{20}{80} \times 100\% = 0.25 \times 100\% = 25\%$

Answer: 25% increase

Interpretation: The price increased by 25%.

Example 4: Percentage Distribution

Problem: In a class of 40 students:

- 12 students prefer math
- 18 students prefer science
- 10 students prefer English

What percentage prefers each subject?

Step 1: Calculate percentage for each subject

Math: $\text{Percentage} = \frac{12}{40} \times 100\% = 0.30 \times 100\% = 30\%$

Science: $\text{Percentage} = \frac{18}{40} \times 100\% = 0.45 \times 100\% = 45\%$

English: $\text{Percentage} = \frac{10}{40} \times 100\% = 0.25 \times 100\% = 25\%$

Step 2: Verify that percentages sum to 100% $30\% + 45\% + 25\% = 100\%$

✓

Subject	Count	Percentage
Math	12	30%
Science	18	45%
English	10	25%
Total	40	100%

Outliers

Outliers are data points that are significantly different from other observations. They can be unusually high or unusually low values that don't fit the general pattern of the data.

- **Definition:** Values that lie outside the expected range of the data.
- **Impact:** Can significantly affect the mean and other statistical measures.
- **Detection Method:** Using the Interquartile Range (IQR) method.

Interquartile Range (IQR) Method

Formulas:

1. **Interquartile Range (IQR):** $IQR = Q3 - Q1$
2. **Lower Fence:** $Lower\ Fence = Q1 - 1.5 \times IQR$
3. **Upper Fence:** $Upper\ Fence = Q3 + 1.5 \times IQR$

Rule: Any value below the lower fence or above the upper fence is considered an outlier.

Example: Detecting Outliers

Dataset: Test scores of 15 students (sorted): 35, 45, 52, 58, 63, 65, 70, 72, 75, 78, 85, 88, 90, 95, 150

Step 1: Calculate Q1 (25th percentile) $n = 15$
 $L = \frac{25}{100} \times (15 + 1) = 0.25 \times 16 = 4$

Position 4 = 58 $Q1 = 58$

Step 2: Calculate Q3 (75th percentile) $L = \frac{75}{100} \times (15 + 1) = 0.75 \times 16 = 12$

Position 12 = 88 $Q3 = 88$

Step 3: Calculate IQR $IQR = Q3 - Q1 = 88 - 58 = 30$

Step 4: Calculate Lower Fence $Lower\ Fence = Q1 - 1.5 \times IQR = 58 - 1.5 \times 30 = 58 - 45 = 13$

Step 5: Calculate Upper Fence $\text{Upper Fence} = Q3 + 1.5 \times \text{IQR} = 88 + 1.5 \times 30 = 88 + 45 = 133$

Step 6: Identify outliers

- Any value < 13 is an outlier
- Any value > 133 is an outlier

Checking the data:

- $35 > 13$ (not a lower outlier)
- $150 > 133$ (UPPER OUTLIER!)

Outliers detected: 150

Interpretation: The score of 150 is unusually high compared to the rest of the data and is considered an outlier.

Summary Table

Measure	Value
Q1	58
Q3	88
IQR	30
Lower Fence	13
Upper Fence	133
Outliers	150

Box Plot

A **box plot** (or box-and-whisker plot) is a graphical representation of data that shows the distribution based on five key values: minimum, Q1, median (Q2), Q3, and maximum.

- **Components:**
 - **Minimum:** Smallest value (excluding outliers)
 - **Q1 (First Quartile):** 25th percentile
 - **Median (Q2):** 50th percentile (middle line in the box)
 - **Q3 (Third Quartile):** 75th percentile
 - **Maximum:** Largest value (excluding outliers)

- **Outliers:** Plotted as individual points
- **Box:** Extends from Q1 to Q3 (represents the IQR)
- **Whiskers:** Lines extending from the box to the minimum and maximum (excluding outliers)

Five-Number Summary

The **five-number summary** consists of:

1. Minimum (excluding outliers)
2. Q1 (25th percentile)
3. Median (50th percentile)
4. Q3 (75th percentile)
5. Maximum (excluding outliers)

Example: Creating a Box Plot

Dataset: Test scores: 35, 45, 52, 58, 63, 65, 70, 72, 75, 78, 85, 88, 90, 95, 150

From our previous calculations:

- $Q1 = 58$
- $Q2 \text{ (Median)} = 72$
- $Q3 = 88$
- $IQR = 30$
- Lower Fence = 13
- Upper Fence = 133
- Outliers = 150

Step 1: Calculate the Median (Q2)

$$n = 15 \text{ (odd number)} \quad \text{Position} = \frac{15 + 1}{2} = 8$$

$$\text{Value at position 8} = 72 \quad \text{Median (Q2)} = 72$$

Step 2: Identify minimum and maximum (excluding outliers)

- Minimum value (excluding outliers) = 35 (first value ≥ 13)
- Maximum value (excluding outliers) = 95 (last value ≤ 133)
- Outlier = 150 (> 133)

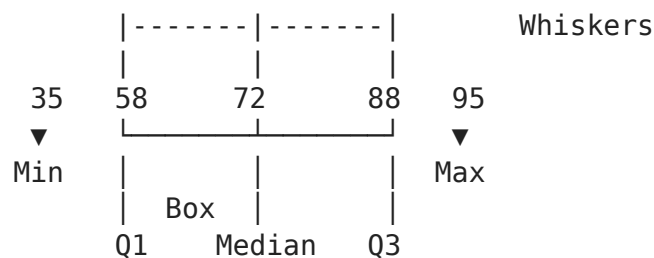
Step 3: Create Five-Number Summary

Statistic	Value
Minimum	35
Q1	58
Median (Q2)	72
Q3	88
Maximum	95
Outlier	150

Step 4: Box Plot Structure

Visual Representation:

Outlier: 150 •



Box Plot Components:

- **Box:** Extends from Q1 (58) to Q3 (88), with a line at the median (72)
- **Left Whisker:** Extends from Q1 (58) down to minimum (35)
- **Right Whisker:** Extends from Q3 (88) up to maximum (95)
- **Outlier:** Point at 150, plotted separately beyond the whisker

Interpretation:

- The box represents the middle 50% of the data (IQR = 30 points)
- The median (72) is close to the center of the box, suggesting fairly symmetric distribution within the IQR
- The value 150 is an outlier, appearing as a separate point
- Most data is concentrated between 58 and 88

Reading a Box Plot

What each part tells you:

1. **Box Width (IQR):** Shows the spread of the middle 50% of data
2. **Median Position:** Shows central tendency and skewness

- If median is in the center of the box → symmetric distribution
 - If median is closer to Q1 → right-skewed distribution
 - If median is closer to Q3 → left-skewed distribution
3. **Whisker Length:** Shows the spread of the outer 50% of data
4. **Outliers:** Individual unusual points that don't follow the pattern

Comparison of All Concepts

Concept	Purpose	Key Formula/Method	Use Case
Percentiles	Show relative position in dataset	$L = \frac{P}{100} \times (n + 1)$	Comparing test scores, salary rankings
Percentages	Express proportion of whole	$\frac{\text{Part}}{\text{Whole}} \times 100\%$	Calculating grades, growth rates, distributions
Outliers	Identify unusual values	$Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$	Data cleaning, quality control, fraud detection
Box Plot	Visualize data distribution	Five-number summary	Comparing distributions, identifying skewness

Complete Example: All Concepts Together

Dataset: Monthly sales (in thousands): 12, 15, 18, 20, 22, 25, 28, 30, 32, 35, 38, 40, 95

Step 1: Calculate Percentiles

$n = 13$

Q1 (25th percentile): $L = \frac{25}{100} \times 14 = 3.5$ Interpolate between position 3 (18) and position 4 (20): $Q1 = 18 + 0.5 \times (20 - 18) = 19$

Q2 (50th percentile/Median): $L = \frac{50}{100} \times 14 = 7$ Position 7 = 28 $Q2 = 28$

Q3 (75th percentile): $L = \frac{75}{100} \times 14 = 10.5$ Interpolate between position 10 (35) and position 11 (38): $Q3 = 35 + 0.5 \times (38 - 35) = 36.5$

Step 2: Calculate IQR and Detect Outliers

$$IQR = Q3 - Q1 = 36.5 - 19 = 17.5$$

$$\text{Lower Fence} = 19 - 1.5 \times 17.5 = 19 - 26.25 = -7.25$$

$$\text{Upper Fence} = 36.5 + 1.5 \times 17.5 = 36.5 + 26.25 = 62.75$$

Outliers: 95 > 62.75 (outlier detected!)

Step 3: Calculate Percentage Above Average

$$\text{Mean} = \frac{12 + 15 + \dots + 95}{13} = \frac{410}{13} = 31.54$$

Values above 31.54: 32, 35, 38, 40, 95 (5 values)

$$\text{Percentage} = \frac{5}{13} \times 100\% = 38.46\%$$

Step 4: Five-Number Summary for Box Plot

Statistic	Value
Minimum	12
Q1	19
Median	28
Q3	36.5
Maximum	40
Outlier	95

Final Summary:

- 25% of months had sales below \$19,000
- 50% of months had sales below \$28,000 (median)
- 75% of months had sales below \$36,500
- 38.46% of months had sales above the average
- \$95,000 in sales is an outlier (unusually high)

Key Insights

Percentiles:

- Divide data into 100 equal parts
- Useful for understanding relative position

- Quartiles (Q1, Q2, Q3) are the most common percentiles used

Percentages:

- Express parts of a whole as fractions of 100
- Essential for comparisons and proportions
- Always verify that percentages sum to 100% when dealing with distributions

Outliers:

- Can significantly affect statistical measures
- Should be investigated (may indicate errors or important exceptions)
- IQR method provides objective criteria for detection

Box Plots:

- Provide visual summary of data distribution
- Show central tendency, spread, and outliers simultaneously
- Excellent for comparing multiple datasets
- Help identify skewness and symmetry in data

```
In [ ]: #Disruption of data
df1.describe()
```

```
Out[ ]:
```

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

```
In [ ]: #Min And Max
np.percentile(df1['bmi'], 0), np.percentile(df1['bmi'], 100)
```

```
Out[ ]: (np.float64(15.96), np.float64(53.13))
```

```
In [ ]: # Quartiles: Q1, Q2, Q3
```

```

# Method 1: Using NumPy
Q1 = np.percentile(df1['bmi'], 25)
Q2 = np.percentile(df1['bmi'], 50)    # Median
Q3 = np.percentile(df1['bmi'], 75)

print(f"NumPy -> Q1: {Q1}, Q2: {Q2}, Q3: {Q3}")

# Method 2: Using pandas
Q1 = df1['bmi'].quantile(0.25)
Q2 = df1['bmi'].quantile(0.50)        # Median
Q3 = df1['bmi'].quantile(0.75)

print(f"Pandas -> Q1: {Q1}, Q2: {Q2}, Q3: {Q3}")

```

NumPy -> Q1: 26.29625, Q2: 30.4, Q3: 34.69375
Pandas -> Q1: 26.29625, Q2: 30.4, Q3: 34.69375

```

In [ ]: # IQR
IQR = Q3 - Q1

print(f"Q1: {Q1}")
print(f"Q3: {Q3}")
print(f"IQR: {IQR}")

```

Q1: 26.29625
Q3: 34.69375
IQR: 8.3975

```

In [ ]: # Lower and Upper Bound
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print(f"Lower Bound: {lower_bound}")
print(f"Upper Bound: {upper_bound}")

```

Lower Bound: 13.7
Upper Bound: 47.290000000000006

Covariance and Correlation

Simple guide to understand how two variables are related.

What is Covariance?

Covariance tells if two things change together.

Three Cases:

- **Positive (+):** Both go up together (study more = score more)
 - **Negative (-):** One up, one down (temperature up = jacket sales down)
 - **Zero (0):** No relationship (shoe size and IQ)
-

Covariance Formula

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

Steps:

- Find average of X and Y
 - Subtract average from each value
 - Multiply the differences
 - Add them up and divide by (n-1)
-

Example 1: Study Hours and Scores

Data:

Student	Hours (X)	Score (Y)
1	2	65
2	3	70
3	4	75
4	5	80
5	6	85
6	7	88
7	8	92
8	9	95

Step 1: Find Averages

Average hours = (2+3+4+5+6+7+8+9) ÷ 8 = **5.5**

Average score = (65+70+75+80+85+88+92+95) ÷ 8 = **81.25**

Step 2: Calculate Differences and Products

Student	X	Y	X-5.5	Y-81.25	Product
1	2	65	-3.5	-16.25	56.88
2	3	70	-2.5	-11.25	28.13
3	4	75	-1.5	-6.25	9.38
4	5	80	-0.5	-1.25	0.63
5	6	85	0.5	3.75	1.88
6	7	88	1.5	6.75	10.13
7	8	92	2.5	10.75	26.88
8	9	95	3.5	13.75	48.13

Step 3: Sum All Products

Total = $56.88 + 28.13 + 9.38 + 0.63 + 1.88 + 10.13 + 26.88 + 48.13 = \mathbf{182}$

Step 4: Calculate Covariance

Cov = $182 \div (8-1) = 182 \div 7 = \mathbf{26}$

Result: Covariance = +26

Meaning: Positive value means study hours and scores go up together!

Example 2: Temperature and Hot Chocolate

Data:

Day	Temp °C (X)	Sales (Y)
1	25	15
2	22	20
3	18	28
4	15	35
5	12	42
6	10	48
7	8	52

Day	Temp °C (X)	Sales (Y)
8	5	58
9	3	62
10	2	65

Step 1: Find Averages

Average temp = $120 \div 10 = \mathbf{12^{\circ}\text{C}}$

Average sales = $425 \div 10 = \mathbf{42.5}$

Step 2: Calculate Products

Day	X	Y	X-12	Y-42.5	Product
1	25	15	13	-27.5	-357.5
2	22	20	10	-22.5	-225
3	18	28	6	-14.5	-87
4	15	35	3	-7.5	-22.5
5	12	42	0	-0.5	0
6	10	48	-2	5.5	-11
7	8	52	-4	9.5	-38
8	5	58	-7	15.5	-108.5
9	3	62	-9	19.5	-175.5
10	2	65	-10	22.5	-225

Step 3: Sum and Calculate

Total = -1250

Cov = $-1250 \div 9 = \mathbf{-138.89}$

Result: Covariance = -138.89

Meaning: Negative value means opposite direction. Hot weather = less hot chocolate!

Problem with Covariance

Issue: Hard to know if the number is big or small!

- Is 26 strong? Weak? Can't tell!
- Is -138.89 stronger than 26? Can't compare!

Solution: Use Correlation!

What is Correlation?

Correlation = Covariance made easy to understand

- Always between **-1** and **+1**
- Shows strength AND direction
- Easy to compare

How to Read Correlation:

- **+1.0** = Perfect positive (exact match)
 - **+0.7 to +1.0** = Strong positive
 - **+0.3 to +0.7** = Moderate positive
 - **0 to +0.3** = Weak positive
 - **0** = No relationship
 - **-0.3 to 0** = Weak negative
 - **-0.7 to -0.3** = Moderate negative
 - **-1.0 to -0.7** = Strong negative
 - **-1.0** = Perfect negative (exact opposite)
-

Correlation Formula

$$r = \frac{\text{Cov}(X, Y)}{S_X \times S_Y}$$

Where:

- r = correlation
- $\text{Cov}(X, Y)$ = covariance
- S_X = standard deviation of X
- S_Y = standard deviation of Y

Standard Deviation Formula:

$$S = \sqrt{\frac{\sum (X - \bar{X})^2}{n-1}}$$

Example 3: Study Hours Correlation

Using same data from Example 1

We already have:

- $Cov(X,Y) = 26$
- Average $X = 5.5$
- Average $Y = 81.25$

Step 1: Calculate $(X - \bar{X})^2$

Student	X	X-5.5	$(X-5.5)^2$
1	2	-3.5	12.25
2	3	-2.5	6.25
3	4	-1.5	2.25
4	5	-0.5	0.25
5	6	0.5	0.25
6	7	1.5	2.25
7	8	2.5	6.25
8	9	3.5	12.25

Sum = **42**

Step 2: Calculate $(Y - \bar{Y})^2$

Student	Y	Y-81.25	$(Y-81.25)^2$
1	65	-16.25	264.06
2	70	-11.25	126.56
3	75	-6.25	39.06
4	80	-1.25	1.56
5	85	3.75	14.06
6	88	6.75	45.56

Student	Y	Y-81.25	\$(Y-81.25)^2\$
7	92	10.75	115.56
8	95	13.75	189.06

Sum = **795.5**

Step 3: Calculate Standard Deviations

$$S_X = \sqrt{42 \div 7} = \sqrt{6} = 2.45$$

$$S_Y = \sqrt{795.5 \div 7} = \sqrt{113.64} = 10.66$$

Step 4: Calculate Correlation

$$r = \frac{26}{2.45 \times 10.66} = \frac{26}{26.12} = 0.995$$

Result: $r = 0.995 \approx 1.0$

Meaning:

- Almost perfect positive correlation!
- Study hours **STRONGLY** predict scores
- Very strong relationship

Example 4: Temperature Correlation

Using data from Example 2

We have:

- $\text{Cov}(X,Y) = -138.89$
- Average $X = 12$
- Average $Y = 42.5$

Step 1: Calculate Standard Deviations

For X (temperature):

Day	X	X-12	\$(X-12)^2\$
1	25	13	169
2	22	10	100

Day	X	X-12	\$(X-12)^2\$
3	18	6	36
4	15	3	9
5	12	0	0
6	10	-2	4
7	8	-4	16
8	5	-7	49
9	3	-9	81
10	2	-10	100

Sum = **564**

$$S_X = \sqrt{564 \div 9} = \sqrt{62.67} = 7.92$$

For Y (sales):

Sum of $(Y-42.5)^2 = \mathbf{2776.5}$

$$S_Y = \sqrt{2776.5 \div 9} = \sqrt{308.5} = 17.56$$

Step 2: Calculate Correlation

$$r = \frac{-138.89}{7.92 \times 17.56} = \frac{-138.89}{139.08} = -0.999$$

Result: $r = -0.999 \approx -1.0$

Meaning:

- Almost perfect negative correlation!
- Temperature and sales move exactly opposite
- Very predictable inverse relationship

Comparison Table

Feature	Covariance	Correlation
Range	$-\infty$ to $+\infty$	-1 to +1
Easy to understand?	No	Yes
Shows direction?	Yes	Yes
Shows strength?	No	Yes

Feature	Covariance	Correlation
Affected by units?	Yes	No
Can compare datasets?	No	Yes

Quick Guide

When Positive (+):

- Both variables increase together
- Example: Study hours UP = Scores UP

When Negative (-):

- Variables move opposite
- Example: Temperature UP = Hot chocolate DOWN

Strength Guide:

- **0 to 0.3** = Weak
 - **0.3 to 0.7** = Moderate
 - **0.7 to 1.0** = Strong
-

Important Warnings

1. Correlation $\neq$ Causation

- High correlation doesn't mean one causes the other
- Example: Ice cream sales and drowning both increase in summer (both caused by hot weather)

2. Only Shows Linear Relationships

- Curved patterns might show $r = 0$ even if related
- Always plot your data first!

3. Watch for Outliers

- One extreme value can change everything
 - Check your data for unusual points
-

Summary

Covariance:

- Shows IF variables are related
- Shows direction (+ or -)
- Hard to interpret the number

Correlation:

- Shows HOW STRONG the relationship is
- Shows direction (+ or -)
- Easy to interpret (always -1 to +1)
- Use this for final conclusions!

Key Formulas:

Covariance:
$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

Correlation:
$$r = \frac{\text{Cov}(X, Y)}{S_X \times S_Y}$$

Practice Problems

Problem 1: Calculate covariance and correlation:

- X: 1, 2, 3, 4, 5
- Y: 10, 20, 30, 40, 50

Problem 2: If correlation between price and sales is -0.85, what does this mean?

Problem 3: Which is stronger: $r = 0.6$ or $r = -0.8$?

Remember:

- Use **Covariance** to calculate
- Use **Correlation** to interpret!

Probability

Terminology - Five Terms

1. Random Experiment

An experiment is called **random experiment** if it satisfies the following two conditions:

- (i) It has more than one possible outcome.
- (ii) It is not possible to predict the outcome in advance.

Real-World Examples:

- **Tossing a coin** - Can get heads or tails, cannot predict which one
 - **Rolling a die** - Can get 1, 2, 3, 4, 5, or 6, outcome is uncertain
 - **Drawing a card from deck** - Can get any of 52 cards, unpredictable
-

2. Trial

Trial refers to a **single execution** of a random experiment. Each trial produces an outcome.

Real-World Examples:

- **One coin flip** - Each time you flip is one trial
 - **One die roll** - Rolling the die once is one trial
 - **Checking one product in factory** - Testing if it's defective or not is one trial
-

3. Outcome

Outcome refers to a **single possible result** of a trial.

Real-World Examples:

- **Coin toss** - Head or Tail (2 outcomes)
 - **Die roll** - Getting a 4 (one specific outcome)
 - **Customer response** - "Yes" or "No" (each is an outcome)
-

4. Sample Space

Sample Space of a random experiment is the **set of all possible outcomes** that can occur. Generally, one random experiment will have one set of sample space.

Notation: S or Ω (Omega)

Real-World Examples:

- **Coin toss:** $S = \{\text{Head, Tail}\}$
 - **Die roll:** $S = \{1, 2, 3, 4, 5, 6\}$
 - **Choosing ice cream flavor:** $S = \{\text{Vanilla, Chocolate, Strawberry, Mango}\}$
-

5. Event

Event is a **specific set of outcomes** from a random experiment or process. Essentially, it's a subset of the sample space.

Real-World Examples:

- **Getting an even number** on die roll: $E = \{2, 4, 6\}$
 - **Getting heads** in coin toss: $E = \{\text{Head}\}$
 - **Student scoring above 80%:** $E = \{\text{all scores} > 80\}$
-

Examples

Example 1: Tossing a Coin

- **Random Experiment:** Tossing a coin
 - **Trial:** One coin flip
 - **Outcome:** Head (H) or Tail (T)
 - **Sample Space:** $S = \{H, T\}$
 - **Event:** Getting Head = $\{H\}$
-

Example 2: Rolling a Die

- **Random Experiment:** Rolling a six-sided die
- **Trial:** One roll
- **Outcome:** 1, 2, 3, 4, 5, or 6

- **Sample Space:** $S = \{1, 2, 3, 4, 5, 6\}$
 - **Event:** Getting odd number = $\{1, 3, 5\}$
-

Example 3: Drawing Cards

- **Random Experiment:** Drawing one card from deck
 - **Trial:** One card drawn
 - **Outcome:** Any one of 52 cards
 - **Sample Space:** $S = \{\text{All 52 cards}\}$
 - **Event:** Getting a Heart = $\{13 \text{ heart cards}\}$
-

Types of Events

1. Simple Event

An event with **only one outcome**.

Real-World Examples:

- Getting exactly 5 on a die roll
 - Drawing the Ace of Spades
 - Getting exactly Heads on coin toss
-

2. Compound Event

An event with **two or more outcomes**.

Real-World Examples:

- Getting an even number on die (2, 4, or 6)
 - Drawing any King from deck (4 possible)
 - Rain on Monday OR Tuesday
-

3. Certain Event

An event that **always happens** (probability = 1).

Real-World Examples:

- Getting a number less than 7 when rolling a die

- Sun rising tomorrow
 - Getting a card from deck when you draw one
-

4. Impossible Event

An event that **never happens** (probability = 0).

Real-World Examples:

- Getting 7 on a standard six-sided die
 - Getting both heads and tails on single coin flip
 - Drawing a yellow card from standard deck
-

5. Complementary Event

The event that **does NOT occur**. If A is an event, then A' (or A^c) is complement.

Formula: $P(A') = 1 - P(A)$

Real-World Examples:

- Event A = Rain, Event A' = No Rain
 - Event A = Pass exam, Event A' = Fail exam
 - Event A = Even number on die, Event A' = Odd number
-

6. Mutually Exclusive Events

Two events that **cannot happen at the same time**.

Real-World Examples:

- Getting Heads AND Tails on same coin flip
 - A student passing AND failing the same exam
 - Turning left AND right at same intersection
-

7. Exhaustive Events

Events that **cover all possible outcomes** of sample space.

Real-World Examples:

- {Even numbers, Odd numbers} on die roll
 - {Pass, Fail} for an exam
 - {Rain, No Rain} for weather
-

What is Probability?

Probability is a measure of how likely an event is to occur.

Formula:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Range: $0 \leq P(A) \leq 1$

- **$P(A) = 0$** → Impossible event
 - **$P(A) = 1$** → Certain event
 - **$P(A) = 0.5$** → Equal chance
-

Real-World Examples:

Example 1: Coin Toss

- $P(\text{Head}) = 1/2 = 0.5 = 50\%$

Example 2: Die Roll

- $P(\text{Getting } 3) = 1/6 \approx 0.167 = 16.7\%$
- $P(\text{Even number}) = 3/6 = 0.5 = 50\%$

Example 3: Drawing Cards

- $P(\text{Drawing an Ace}) = 4/52 = 1/13 \approx 0.077 = 7.7\%$
-

Random Variable

In probability theory, a random variable is a function that maps the outcomes of a random experiment (called the sample space) to a set of real numbers.

Input

The input to the function is an outcome from the sample space of a random

process.

Output

The output of the function is a real number that we assign to each possible outcome.

Simple Meaning

Random Variable = Function

Input → Sample Space

Outcome → Real Number

Example 1: Rolling a Die

Sample Space:

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let

Y = Score on die roll

Mapping:

$$1 \rightarrow 1 \quad 2 \rightarrow 2 \quad 3 \rightarrow 3 \quad 4 \rightarrow 4 \quad 5 \rightarrow 5 \quad 6 \rightarrow 6$$

So,

$$Y \in \{1, 2, 3, 4, 5, 6\}$$

Example 2: Tossing 3 Coins

Sample Space:

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Let

X = Number of Heads

Mapping:

$$HHH \rightarrow 3 \quad HHT \rightarrow 2 \quad HTH \rightarrow 2 \quad HTT \rightarrow 1 \quad THH \rightarrow 2 \quad THT \rightarrow 1 \quad TTH \rightarrow 1 \quad TTT \rightarrow 0$$

So,

$$X \in \{0, 1, 2, 3\}$$

Important Points

A random variable is actually a function

It assigns numerical values to outcomes

It depends on how we define it

Different definitions give different random variables

Discrete and Continuous Variables

Discrete Random Variable

Takes **specific, countable values** (usually integers).

Characteristics:

- Can list all possible values
- Often result of counting

Real-World Examples:

- Number of students in class (20, 21, 22...)
 - Number of emails received (0, 1, 2, 3...)
 - Number of cars sold in a day (0, 1, 2...)
-

Continuous Random Variable

Can take **any value within a range** (infinite possibilities).

Characteristics:

- Cannot list all values
- Often result of measuring

Real-World Examples:

- Height of students (165.5 cm, 170.2 cm, 175.8 cm...)
 - Time taken to complete task (12.5 min, 15.3 min...)
 - Temperature in a city (25.3°C, 30.7°C...)
-

Probability Distribution of a Random Variable

A **Probability Distribution** shows all possible values a random variable can take and their probabilities.

For Discrete Variables: Probability Mass Function (PMF)

For Continuous Variables: Probability Density Function (PDF)

Example: Rolling a Die

X (Outcome)	P(X)
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

Sum of all probabilities = 1

Real-World Examples:

Example 1: Number of Defective Items

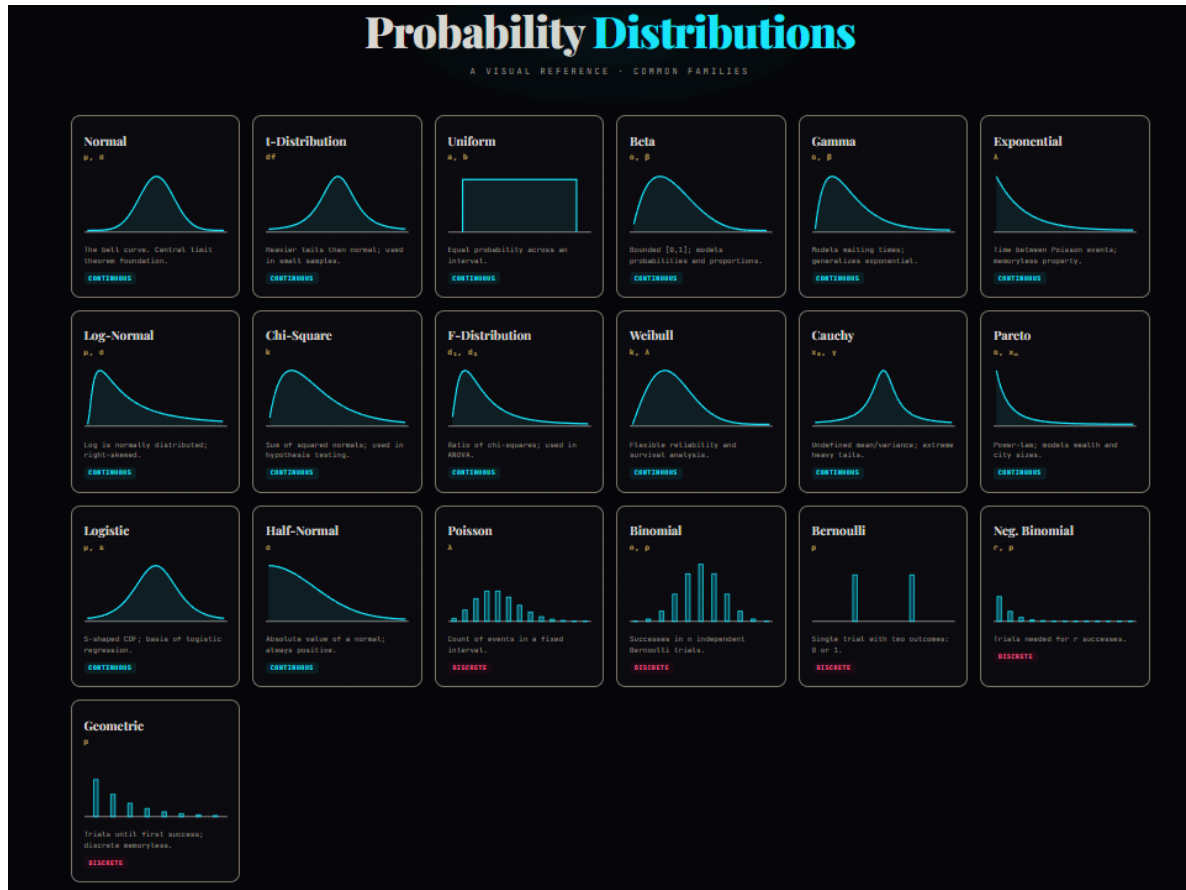
- $X = 0$ defects, $P(X=0) = 0.7$
- $X = 1$ defect, $P(X=1) = 0.2$
- $X = 2$ defects, $P(X=2) = 0.1$

Example 2: Customer Satisfaction (1-5 scale)

- Very Poor (1): 5%
- Poor (2): 10%
- Average (3): 30%
- Good (4): 35%
- Excellent (5): 20%

Example 3: Number of Goals in Football Match

- 0 goals: 20%
- 1 goal: 35%
- 2 goals: 25%
- 3+ goals: 20%



Venn Diagrams in Probability

Venn Diagrams visually show relationships between events.

Key Operations:

- **Union ($A \cup B$):** Events in A OR B or both
- **Intersection ($A \cap B$):** Events in BOTH A AND B
- **Complement (A'):** Events NOT in A

Visual Guide to Venn Diagrams:

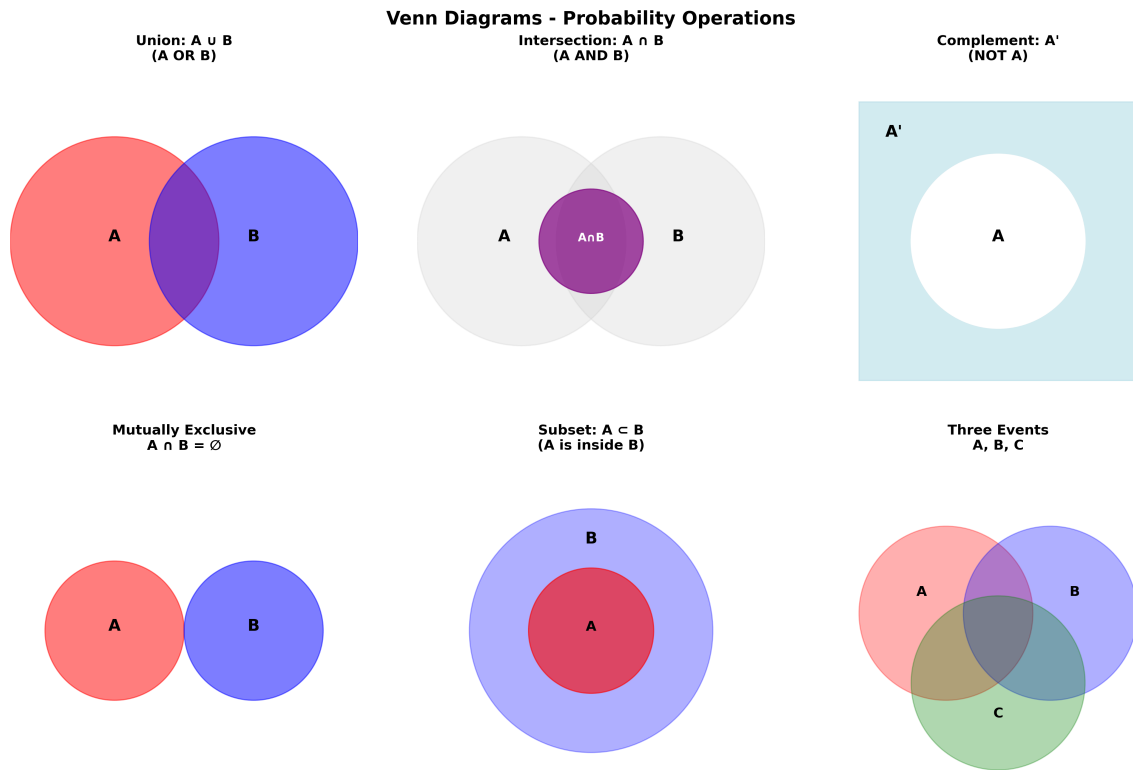


Diagram Explanations:

1. **Union ($A \cup B$)** - All shaded area represents A OR B or both
2. **Intersection ($A \cap B$)** - Purple area represents A AND B (overlap)
3. **Complement (A')** - Blue shaded area represents everything NOT in A
4. **Mutually Exclusive** - No overlap, events cannot happen together
5. **Subset ($A \subset B$)** - All of A is contained within B
6. **Three Events** - Shows how three events can overlap

Example 1: Student Activities

- A = Students who play cricket = 40 students
- B = Students who play football = 35 students
- $A \cap B$ = Play both sports = 15 students
- $A \cup B$ = Play at least one sport = $40 + 35 - 15 = 60$ students

Example 2: Product Features

- A = Products with WiFi = 200 products
- B = Products with Bluetooth = 150 products
- $A \cap B$ = Have both features = 100 products

Example 3: Customer Demographics

- A = Customers aged 18-30 = 500 customers
 - B = Customers who bought online = 600 customers
 - $A \cap B$ = Young online shoppers = 300 customers
 - $A \cup B$ = Young OR online buyers = $500 + 600 - 300 = 800$ customers
-

Contingency Tables in Probability

Contingency Table (Cross-tabulation) shows frequency distribution of two or more categorical variables.

Example 1: Gender and Product Preference

	Product A	Product B	Total
Male	40	60	100
Female	70	30	100
Total	110	90	200

Real-World Examples:

Example 2: Education and Employment

	Employed	Unemployed	Total
Graduate	80	20	100
Non-Graduate	50	50	100

Example 3: Age and Voting

	Voted	Didn't Vote	Total
18-30	300	700	1000
31-50	600	400	1000

	Voted	Didn't Vote	Total
51+	800	200	1000

Joint Probability

Joint Probability is the probability of **two events occurring together**.

Notation: $P(A \cap B)$ or $P(A, B)$

If

$P(A)$ = Probability of getting Head

$P(B)$ = Probability of getting a 6 on dice

Then $P(A \cap B)$ = Probability of getting Head and 6 together (when tossing a coin and rolling a dice at the same time).

If events are independent:

Formula: $P(A \cap B) = P(A) \times P(B)$

Real-World Examples:

Example 1: Drawing Cards

- $P(\text{King AND Heart}) = P(\text{King of Hearts}) = 1/52$

Example 2: Weather and Traffic

- $P(\text{Rain AND Heavy Traffic}) = 0.3 \times 0.6 = 0.18$

Example 3: Customer Purchase

- $P(\text{Buy Laptop AND Buy Mouse}) = 0.4 \times 0.7 = 0.28$

Marginal Probability

Marginal Probability is the probability of a **single event** occurring, regardless of other events.

Found by summing joint probabilities across rows or columns in contingency table.

Real-World Examples:

Using the Gender-Product table:

Example 1:

- $P(\text{Male}) = 100/200 = 0.5$ (marginal probability)

Example 2:

- $P(\text{Product A}) = 110/200 = 0.55$ (marginal probability)

Example 3: Student Performance

- $P(\text{Pass}) = \text{Total students who passed} / \text{Total students}$
- (Regardless of gender, course, or other factors)

```
In [ ]: import numpy as np
import pandas as pd

df2=pd.read_csv('/content/Titanic-Dataset.csv')
```

```
In [ ]: df2.head()
```

```
Out[ ]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450

```
In [ ]: #make contengency table
pd.crosstab(df2['Survived'],df2['Pclass'])
```

```
Out[ ]:
```

Pclass	1	2	3
Survived			
0	80	97	372
1	136	87	119

```
In [ ]: 80/891
```

```
Out[ ]: 0.08978675645342311
```

```
In [ ]: # joint probability of Survived and Pclass
pd.crosstab(df2['Survived'], df2['Pclass'], normalize='all')
```

```
Out[ ]:
```

Pclass	1	2	3
Survived			
0	0.089787	0.108866	0.417508
1	0.152637	0.097643	0.133558

```
In [ ]: (80+136)/891
```

```
Out[ ]: 0.24242424242424243
```

```
In [ ]: # Marginal probability of Survived and Pclass
pd.crosstab(df2['Survived'], df2['Pclass'], normalize='all', margins=True)
```

```
Out[ ]:
```

	Pclass	1	2	3	All
Survived					
0	0.089787	0.108866	0.417508	0.616162	
1	0.152637	0.097643	0.133558	0.383838	
All	0.242424	0.206510	0.551066	1.000000	

Conditional Probability

Conditional Probability is the probability of event A occurring **given that** event B has already occurred.

Notation: $P(A|B)$ - read as "probability of A given B"

Formula: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

where $P(B) > 0$

Real-World Examples:

Example 1: Medical Testing

- $P(\text{Disease} | \text{Positive Test})$ = Probability of having disease GIVEN test is positive

Example 2: Product Quality

- $P(\text{Defective} | \text{Factory A})$ = Probability product is defective GIVEN it came from Factory A

Example 3: Weather

- $P(\text{Traffic Jam} | \text{Rain})$ = Probability of traffic jam GIVEN it's raining
-

Intuition of Conditional Probability Formula

Why do we divide by $P(B)$?

When we know B has occurred, **B becomes our new sample space.**

Think of it this way:

Original Sample Space: All possible outcomes

New Sample Space: Only outcomes where B occurred

We want: What fraction of B's outcomes also include A?

Real-World Examples:

Example 1: Email Classification

- 1000 emails total
- 300 are spam (B)
- 150 spam emails contain word "offer" ($A \cap B$)
- $P(\text{Contains "offer"} \mid \text{Spam}) = 150/300 = 0.5 = 50\%$

Example 2: Customer Behavior

- 500 website visitors
- 200 added items to cart (B)
- 80 who added items actually purchased ($A \cap B$)
- $P(\text{Purchase} \mid \text{Added to Cart}) = 80/200 = 0.4 = 40\%$

Example 3: Disease Diagnosis

- 10,000 people tested
- 100 have disease (B)
- 95 with disease tested positive ($A \cap B$)
- $P(\text{Positive Test} \mid \text{Disease}) = 95/100 = 0.95 = 95\%$

Two fair six-sided dice are rolled. What is the conditional probability that the sum of the numbers rolled is 7, given that the first die shows an odd number?

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Independent vs. Dependent vs. Mutually Exclusive Events

1. Independent Events

Two events are **independent** if occurrence of one does NOT affect the other.

Test: $P(A|B) = P(A)$

Formula: $P(A \cap B) = P(A) \times P(B)$

Real-World Examples:

- Tossing coin twice - First toss doesn't affect second
- Rolling two dice - Result of one doesn't affect other
- Choosing random student and weather being sunny - Unrelated

2. Dependent Events

Two events are **dependent** if occurrence of one DOES affect the other.

Test: $P(A|B) \neq P(A)$

Real-World Examples:

- Drawing two cards WITHOUT replacement - First draw affects second
 - $P(\text{Rain tomorrow} | \text{Rain today}) > P(\text{Rain tomorrow})$ - Weather related
 - Studying hard and passing exam - Related events
-

3. Mutually Exclusive Events

Two events are **mutually exclusive** if they CANNOT occur at the same time.

Test: $P(A \cap B) = 0$

Formula: $P(A \cup B) = P(A) + P(B)$

Real-World Examples:

- Getting Heads AND Tails on single coin flip - Impossible
 - Passing AND Failing same exam - Can't happen together
 - Turning left AND right at intersection - Mutually exclusive
-

Comparison Table:

Property	Independent	Dependent	Mutually Exclusive
Can occur together?	Yes	Yes	No
One affects other?	No	Yes	N/A
$P(A \cap B) = ?$	$P(A) \times P(B)$	$P(A) \times P(B A)$	0
$P(A B) = ?$	$P(A)$	$\neq P(A)$	0

Common Confusion:

Mutually Exclusive $\neq$ Independent

In fact, if events are mutually exclusive, they are DEPENDENT!

Why? If A occurs, it tells us B cannot occur → They affect each other!

Bayes' Theorem

Bayes' Theorem helps us update probabilities based on new evidence.

Formula:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Components:

- **P(A|B):** Posterior probability (what we want to find)
 - **P(B|A):** Likelihood (probability of evidence given hypothesis)
 - **P(A):** Prior probability (initial belief)
 - **P(B):** Marginal probability (total probability of evidence)
-

Real-World Examples:

Example 1: Medical Diagnosis

Given:

- 1% of population has disease [$P(\text{Disease}) = 0.01$]
- Test is 95% accurate if you have disease [$P(\text{Positive}|\text{Disease}) = 0.95$]
- Test is 90% accurate if you don't have disease [$P(\text{Negative}|\text{No Disease}) = 0.90$]

Question: If test is positive, what's probability you have disease?

Solution:

$$P(\text{Disease}|\text{Positive}) = (0.95 \times 0.01) / [(0.95 \times 0.01) + (0.10 \times 0.99)]$$

$$= 0.0095 / [0.0095 + 0.099]$$

$$= 0.0095 / 0.1085$$

$$= \mathbf{0.0875 \text{ or } 8.75\%}$$

Interpretation: Even with positive test, only 8.75% chance of disease because disease is rare!

Example 2: Spam Email Detection

Given:

- 30% of emails are spam [$P(\text{Spam}) = 0.3$]
- 80% of spam emails contain "free" [$P(\text{"free"}|\text{Spam}) = 0.8$]
- 10% of legitimate emails contain "free" [$P(\text{"free"}|\text{Not Spam}) = 0.1$]

Question: Email contains "free", what's probability it's spam?

Solution:

$$P(\text{Spam}|\text{"free"}) = (0.8 \times 0.3) / [(0.8 \times 0.3) + (0.1 \times 0.7)]$$

$$= 0.24 / [0.24 + 0.07]$$

$$= 0.24 / 0.31$$

$$= \mathbf{0.774 \text{ or } 77.4\%}$$

Example 3: Criminal Investigation

Given:

- 2% of people in area match description [$P(\text{Match}) = 0.02$]
- 90% chance person at scene left evidence [$P(\text{Evidence}|\text{Guilty}) = 0.9$]
- 5% chance random person has similar evidence [$P(\text{Evidence}|\text{Not Guilty}) = 0.05$]

Question: Person matches description and has evidence, what's probability they're guilty?

Solution:

$$P(\text{Guilty}|\text{Evidence}) = (0.9 \times 0.02) / [(0.9 \times 0.02) + (0.05 \times 0.98)]$$

$$= 0.018 / [0.018 + 0.049]$$

$$= 0.018 / 0.067$$

$$= \mathbf{0.269 \text{ or } 26.9\%}$$

Interpretation: Despite matching evidence, still low probability because many innocent people match description!

Summary

Key Concepts:

- ✓ **Random Experiment:** Unpredictable outcomes
- ✓ **Probability:** Measure of likelihood (0 to 1)
- ✓ **Random Variables:** Numerical outcomes of experiments
- ✓ **Distributions:** How probabilities are spread
- ✓ **Joint Probability:** Two events together
- ✓ **Conditional Probability:** One event given another
- ✓ **Bayes' Theorem:** Update beliefs with evidence

Remember:

- Always check if events are independent or dependent
 - Mutually exclusive events cannot occur together
 - Conditional probability changes the sample space
 - Bayes' theorem helps reverse conditional probabilities
-

Practice Problems

Problem 1: If $P(A) = 0.6$, $P(B) = 0.5$, and A and B are independent, find $P(A \cap B)$.

Problem 2: In a class, 60% students passed Math, 70% passed English, and 50% passed both. If a student passed Math, what's the probability they passed English?

Problem 3: A factory has 3 machines. Machine A produces 50% of items (2% defective), Machine B produces 30% (3% defective), Machine C produces 20% (5% defective). If an item is defective, what's probability it came from Machine C? (Use Bayes' Theorem)

 **Keep Practicing!**

Normal Distribution

What is Normal Distribution?

Normal Distribution (also called **Gaussian Distribution** or **Bell Curve**) is the most important probability distribution in statistics.

It describes how data points are **symmetrically distributed** around a central mean value.

Key Characteristics

- Symmetric about the mean
 - Bell-shaped curve
 - **Mean = Median = Mode**
 - Defined by two parameters: μ (**mean**) and σ (**standard deviation**)
 - Total area under the curve = **1 (100%)**
-

Formula

Probability Density Function (PDF):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Where:

- x = random variable
 - μ = mean (center of distribution)
 - σ = standard deviation (spread of distribution)
 - σ^2 = variance
 - e = Euler's number ≈ 2.718
 - $\pi \approx 3.14159$
-

Standard Normal Distribution (Z-distribution)

When $\mu = 0$ and $\sigma = 1$, the formula simplifies to:

$$f(z) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{z^2}{2}}$$

Z-Score Formula (Standardization)

$$Z = \frac{X - \mu}{\sigma}$$

Z-score tells us how many standard deviations a value is away from the mean.

- **Z = 0** → value is exactly at the mean
- **Z = +1** → value is 1 standard deviation above the mean
- **Z = -2** → value is 2 standard deviations below the mean



The Empirical Rule (68 - 95 - 99.7 Rule)

Range	% of Data	Meaning
$\mu \pm 1\sigma$	68.27%	Most common outcomes
$\mu \pm 2\sigma$	95.45%	Typical range
$\mu \pm 3\sigma$	99.73%	Almost all outcomes

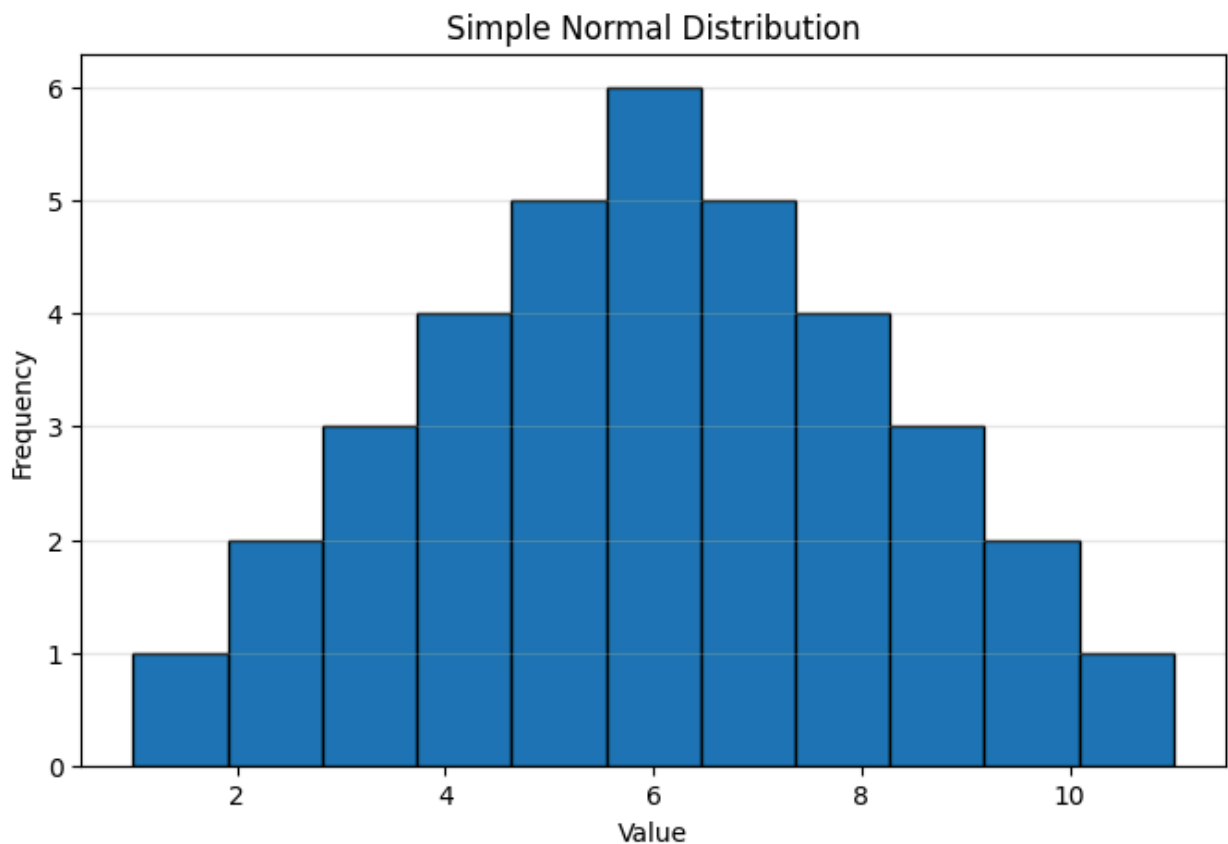
```
In [ ]: import matplotlib.pyplot as plt

# Symmetric bell-shaped data
data = [1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 6, 7, 7, 7]

# Create histogram
plt.figure(figsize=(8,5))
plt.hist(data, bins=11, edgecolor='black')

# Add title and labels
plt.title("Simple Normal Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")

plt.grid(axis='y', alpha=0.3)
plt.show()
```



```
In [ ]: import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# Mean and standard deviation
mu = 0
sigma = 1

# Create values from -4 to 4
x = np.linspace(-4, 4, 1000)

# Calculate normal distribution values
y = norm.pdf(x, mu, sigma)

# Create plot
plt.figure(figsize=(8,5))
plt.plot(x, y, label="Normal Curve")

# Draw mean line
plt.axvline(mu, linestyle="--", label="Mean")

# Draw standard deviation lines
plt.axvline(mu + 1, linestyle=":")
plt.axvline(mu - 1, linestyle=":")
plt.axvline(mu + 2, linestyle=":")
plt.axvline(mu - 2, linestyle=":")
plt.axvline(mu + 3, linestyle=":")
```

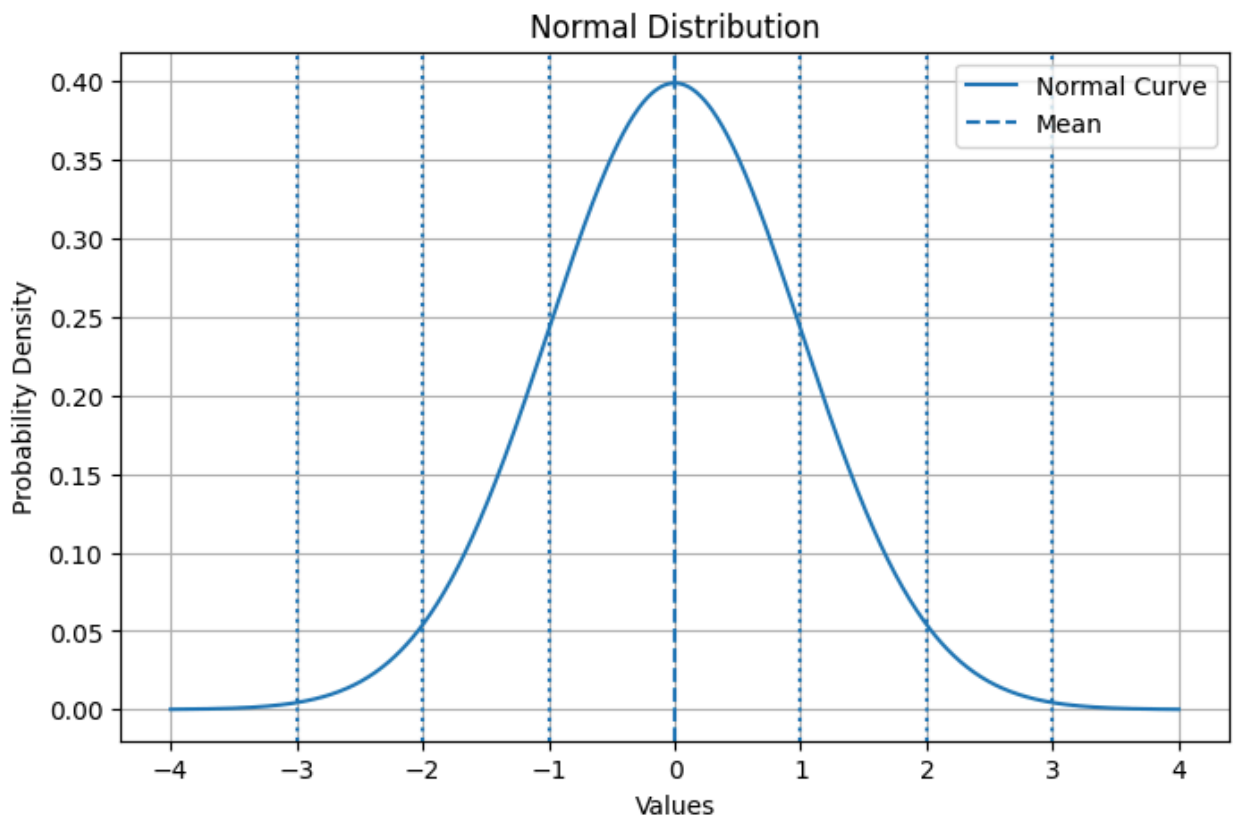


```
plt.axvline(mu - 3, linestyle=":")

# Add title and labels
plt.title("Normal Distribution")
plt.xlabel("Values")
plt.ylabel("Probability Density")

# Show legend and grid
plt.legend()
plt.grid()

# Display graph
plt.show()
```



Central Limit Theorem (CLT)

Central Limit Theorem (CLT) states that if we take a large number of random samples from any population and calculate their means, the distribution of those sample means will be approximately **normal**, even if the original data is **not normally distributed**.

Why CLT is Important

- Real-world data is often **non-normal**
- Allows use of **normal distribution**
- Used in **hypothesis testing**

- Very important in **Machine Learning**
-

Conditions of CLT

1. Samples should be **random**
 2. Sample size should be **large** (generally $n \geq 30$)
 3. Observations must be **independent**
-

What CLT Tells Us

- Mean of sample means = Population mean
 - Standard deviation of sample means:
-

Simple Intuition

- Population data can be of **any shape**
 - Take many samples from population
 - Calculate mean of each sample
 - Plot the sample means
- You get a **bell-shaped (normal) curve**
-

```
In [ ]: import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

pop_data = [np.random.randint(10, 100) for i in range(10000)]
pop_table = pd.DataFrame({"Pop_data": pop_data})
pop_table
```

```
-----
TypeError                                Traceback (most recent call last)
/tmp/ipython-input-4406/3076008585.py in <cell line: 0>()
      4 import pandas as pd
      5
----> 6 pop_data = [np.random.randint(10, 100) for i in range(10000)]
      7 pop_table = pd.DataFrame({"Pop_data": pop_data})
      8 pop_table

TypeError: 'int' object is not callable
```

```
In [ ]: np.mean(pop_data)
```

```
In [ ]: # Create a new figure with size 3x2 inches
plt.figure(figsize=(3,2))

# Draw KDE (Kernel Density Estimation) plot for "Pop_data" column
```

```
sns.kdeplot(x="Pop_data", data=pop_table)

# Display the plot
plt.show()
```

```
In [ ]: # Empty list to store mean of each sample
sample_mean = []

# Loop to create 50 different samples
for no_sample in range(50):

    # Empty list to store one sample of 500 values
    sample_data = []

    # Select 500 random values from population data
    for data in range(500):
        sample_data.append(np.random.choice(pop_data))

    # Calculate mean of current sample and store it
    sample_mean.append(np.mean(sample_data))

# Convert sample means into DataFrame
sample_M = pd.DataFrame({"Sample_Mean": sample_mean})

# Create figure
plt.figure(figsize=(3,2))

# Plot KDE of sample means (sampling distribution)
sns.kdeplot(x="Sample_Mean", data=sample_M)

# Show plot
plt.show()
```

```
In [ ]: np.mean(sample_mean)
```

Hypothesis Testing

Hypothesis testing is a statistical method used to make decisions about a population based on sample data.

Key Terms in Hypothesis Testing

- **Null Hypothesis (H_0):** No effect or no difference
- **Alternative Hypothesis (H_1):** There is an effect or a difference
- **Significance Level (α):** Probability of rejecting a true H_0 (usually 0.05)
- **p-value:** Probability of observing the data if H_0 is true
- **Test Statistic:** Value calculated from sample data

Decision Rule

- If **p-value** $\leq \alpha$ $\rightarrow$ Reject H_0
 - If **p-value** $> \alpha$ $\rightarrow$ Fail to reject H_0
-

Types of Errors

- **Type I Error:** Rejecting a true H_0
 - **Type II Error:** Not rejecting a false H_0
-

Why Hypothesis Testing is Important

- Helps in **data-driven decisions**
 - Used in **model evaluation**
 - Supports **statistical inference**
 - Very important in **Machine Learning experiments**
-

t-Test

A t-test is used to compare **means** when sample size is small and population standard deviation is unknown.

Types of t-Test

- **One-sample t-test:** Sample mean vs population mean
 - **Two-sample t-test:** Compare means of two groups
 - **Paired t-test:** Before-after comparison
-

t-Test Formula

$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

Meaning of Terms (t-Test)

- $\bar{X}$ = Sample mean
 - μ = Population mean
 - s = Sample standard deviation
 - n = Sample size
-

When to Use t-Test

- Data is **continuous**
 - Data is **approximately normal**
 - Sample size is **small ($n < 30$)**
-

Chi-Square Test (χ^2 Test)

Chi-Square test is used to check the **relationship between categorical variables** or to test **goodness of fit**.

Types of Chi-Square Test

- **Chi-Square Test of Independence**
 - **Chi-Square Goodness of Fit Test**
-

Chi-Square Formula

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Meaning of Terms (Chi-Square)

- O = Observed frequency
 - E = Expected frequency
-

When to Use Chi-Square Test

- Data is **categorical**
 - Frequencies are used (counts)
 - Sample size is **large**
 - Expected frequency should be ≥ 5
-

Difference Between t-Test and Chi-Square Test

t-Test

- Used for **numerical data**
- Compares **means**
- Based on **normal distribution**

Chi-Square Test

- Used for **categorical data**
 - Compares **frequencies**
 - Based on **distribution of counts**
-

One-Line Summary

- **t-test:** Compare means
- **Chi-Square test:** Compare categories

```
In [ ]: #t-Test
# Step 1: Import required libraries
import numpy as np
from scipy import stats
```

```
In [ ]: # Step 2: Sample data (e.g., exam marks of students)
sample_data = np.array([65, 70, 72, 68, 75, 78, 74, 69, 71, 73])

# Assumed population mean
population_mean = 70
```

```
In [ ]: # Step 3: Perform one-sample t-test
t_statistic, p_value = stats.ttest_1samp(sample_data, population_mean)

print("t-statistic:", t_statistic)
print("p-value:", p_value)
```

```
In [ ]: # Step 4: Decision based on significance level
alpha = 0.05

if p_value <= alpha:
    print("Reject the Null Hypothesis (H₀)")
else:
    print("Fail to Reject the Null Hypothesis (H₀)")
```

```
In [ ]: #Chi-Square Test (Test of Independence)
# Step 1: Import required libraries
import numpy as np
from scipy.stats import chi2_contingency
```

```
In [ ]: # Step 2: Create contingency table
# Example: Gender vs Product Preference

# Rows -> Gender (Male, Female)
# Columns -> Preference (Yes, No)
data = np.array([
```

```

    [30, 10], # Male
    [20, 40]  # Female
])

```

```

In [ ]: # Step 3: Perform chi-square test
chi2_stat, p_value, dof, expected = chi2_contingency(data)

print("Chi-square statistic:", chi2_stat)
print("p-value:", p_value)
print("Degrees of freedom:", dof)

```

```

In [ ]: # Step 4: Print expected frequencies
print("Expected Frequencies:")
print(expected)

```

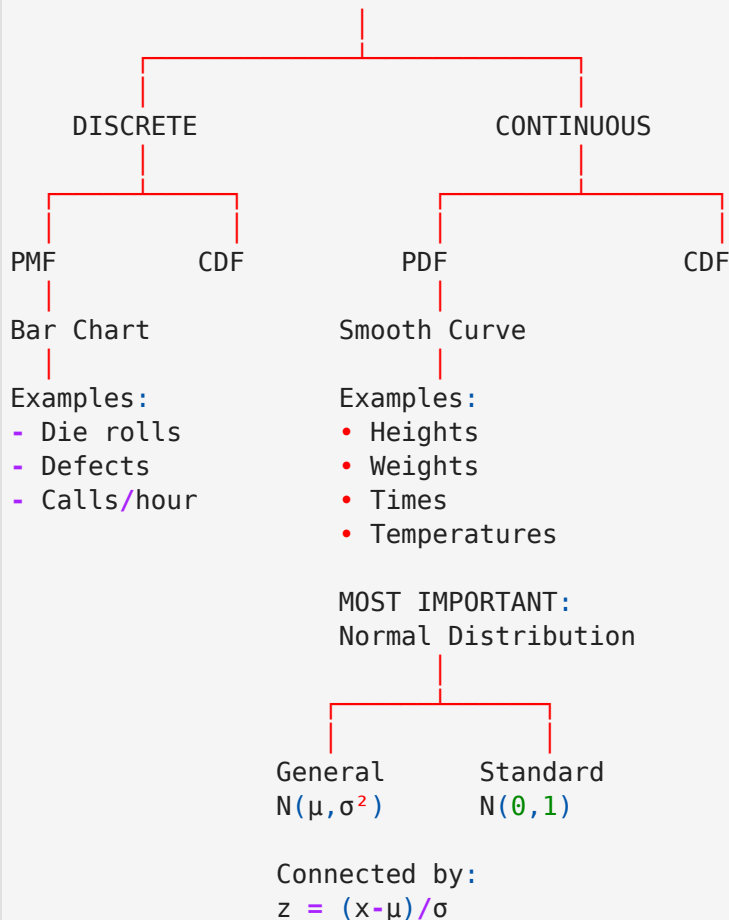
```

In [ ]: # Step 5: Decision based on significance level
alpha = 0.05

if p_value <= alpha:
    print("Reject the Null Hypothesis (H₀)")
else:
    print("Fail to Reject the Null Hypothesis (H₀)")

```

In []: PROBABILITY DISTRIBUTIONS



In []:

In []: